

Color Atlas of Genetics
2nd edition

To my wife, Mary

Color Atlas of Genetics

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Second edition, enlarged and revised

With 194 color plates by Jürgen Wirth



Thieme
Stuttgart · New York 2001

*Library of Congress Cataloging-in-Publication
Data*

Passarge, Eberhard.

[Taschenatlas der Genetik. English]
Color atlas of genetics / Eberhard Passarge, –
2nd ed., enl., and rev.
p. ; cm.

Includes bibliographical references and
index.

ISBN 3131003626 – ISBN 0-86577-958-9

1. Genetics – Atlases.

2. Medical genetics – Atlases. I. Title.

[DNLM: 1. Genetics, Medical – Atlases.

2. Genetics, Medical – Handbooks. QZ 17
P286t 2000a]

QH436 P3713 2000

576.5'022'2 – dc21

00-048874

1st German edition 1994

1st English edition 1995

1st French edition 1995

1st Japanese edition 1996

1st Chinese edition 1998

1st Italian edition 1999

1st Turkish edition 2000

© 2001 Georg Thieme Verlag,
Rüdigerstraße 14, D-70469 Stuttgart, Germany
Thieme New York, 333 Seventh Avenue,
New York, N.Y. 10001 U.S.A.

Color plates by Jürgen Wirth, Professor of
Visual Communication, Fachhochschule
Darmstadt

Typesetting by Druckhaus Götz GmbH,
D-71636 Ludwigsburg
(CCS-Textline [Linotronic 630])
Printed in Germany by Appl, Wemding

ISBN 3-13-100362-6 (GTV)

ISBN 0-86577-958-9 (TNY)

1 2 3 4 5

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Preface

Knowledge about genes (genetics) and genomes (genomics) of different organisms continues to advance at a brisk pace. All manifestations of life are determined by genes and their interactions with the environment. A genetic component contributes to the cause of nearly every human disease. More than a thousand diseases result from alterations in single known genes.

Classical genetics, developed during the first half of the last century, and molecular genetics, developed during the second half, have merged into a fascinating scientific endeavor. This has provided both a theoretical foundation and a broad repertoire of methods to explore cellular mechanisms and to understand normal processes and diseases at the molecular level. Deciphering the genomes of many different organisms, including bacteria and plants, by determining the sequence of the individual building blocks—the nucleotide bases of deoxyribonucleic acid (DNA)—will augment our understanding of normal and abnormal functions. The new knowledge holds promise for the design of pharmaceutical compounds aimed at individual requirements. This will pave the way to new approaches to therapy and prevention. Insights are gained into how organisms are related by evolution.

Students in biology and medicine face an enormous task when attempting to acquire the new knowledge and to interpret it within a conceptual framework. Many good textbooks are available (see General References, p. 421). This Color Atlas differs from standard textbooks by using a visual approach to convey important concepts and facts in genetics. It is based on carefully designed color plates, each accompanied by a corresponding explanatory text on the opposite page.

In 1594 Mercator first used the term “atlas” for a collection of maps. Although maps of genes are highly important in genetics, the term atlas in the context of this book refers to illustrations in general. Here they provide the basis for an in-

troductory, hopefully stimulating interest in an exciting field of study.

This second edition has been extensively revised, rewritten, updated, and expanded. A new section on genomics (Part II) has been added. Twenty new plates deal with a variety of topics such as the molecular bases of genetics, regulation and expression of genes, genomic imprinting, mutations, chromosomes, genes predisposing to cancer, ion channel diseases, hearing and deafness, a brief guide to genetic diagnosis, human evolution, and many others. The Chronology of Important Advances in Genetics and the Definitions of Genetic Terms have been updated. As in the first edition, references are included for further reading. Here and in the list of general references, the reader will find access to more detailed information than can be presented in the limited space available. Websites for further information are included.

A single-author book cannot provide all the details on which scientific knowledge is based. However, it can present an individual perspective suitable as an introduction. In making the difficult decisions about which material to include and which to leave out, I have relied on 25 years' experience of teaching medical students at preclinical and clinical levels. I have attempted to emphasize the intersection of theoretical fundamentals and the medical aspects of genetics, taking a broad viewpoint based on the evolution of living organisms.

All the color plates were produced as computer graphics by Jürgen Wirth, Professor of Visual Communication at the Faculty of Design, University of Applied Sciences, Darmstadt. He created the plates from hand drawings, sketches, photographs, and photocopies assembled by the author. I am deeply indebted to Professor Jürgen Wirth for his most skilful work, the pleasant cooperation, and his patience with all of the author's requests. Without him this book would not have been possible.

Essen, November 2000

E. Passarge

Acknowledgements

In updating, revising, and rewriting this second edition, I received invaluable help from many colleagues who generously provided information and advice, photographic material, and other useful suggestions in their areas of expertise: Hans Esche, Essen; Ulrich Langenbeck, Frankfurt; Clemens Müller-Reible, Würzburg; Maximilian Muenke, Bethesda, Maryland; Stefan Mundlos, Berlin; Alfred Pühler, Bielefeld; Gudrun Rappold, Heidelberg; Helga Rehder, Marburg; Hans Hilger Ropers, Berlin; Gerd Scherer, Freiburg; Evelyn Schröck, Bethesda, Maryland; Eric Schulze-Bahr, Münster; Michael Speicher, München; Manfred Stuhmann-Spangenberg, Hannover; Gerd Utermann, Innsbruck; and Douglas C Wallace and Marie Lott, Atlanta.

In addition, the following colleagues at our Department of Human Genetics, University of Essen Medical School, made helpful suggestions: Beate Albrecht, Karin Buiting, Gabriele Gillesen-Kaesbach, Cornelia Hardt, Bernhard Horsthemke, Frank Kaiser, Dietmar Lohmann, Hermann-Josef Lüdecke, Eva-Christina Prott, Maren Runte, Frank Tschentscher, Dagmar Wiczorek, and Michael Zeschnigk.

I thank my wife, Dr. Mary Fetter Passarge, for her careful reading and numerous helpful suggestions. Liselotte Freimann-Gansert and Astrid Maria Noll transcribed the many versions of the text. I am indebted to Dr. Clifford Bergman, Ms Gabriele Kuhn, Mr Gert Krüger, and their co-workers at Thieme Medical Publishers Stuttgart for their excellent work and cooperative spirit.

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Table of Contents (Overview)

Indroduction	1	Immune System	300
Chronology of Important		Origin of Tumors	316
Advances in Genetics	13	Oxygen and Electron Transport	336
		Lysosomes and LDL Receptor	352
Part I. Fundamentals	19	Homeostasis	362
Molecular Basis of Genetics	20	Maintaining Cell and Tissue Shape	374
Prokaryotic Cells and Viruses	84	Mammalian Sex Determination and	
Eukaryotic Cells	104	Differentiation	386
Mitochondrial Genetics	124	Atypical Inheritance Pattern	394
Formal Genetics	132	Karyotype/Phenotype Correlation	400
Chromosomes	170	A Brief Guide to Genetic Diagnosis	406
Regulation and Expression of Genes	204		
		Chromosomal Location of	
Part II. Genomics	233	Monogenic Diseases	410
		General References	421
Part III. Genetics and		Glossary	423
Medicine	263	Index	442
Cell-to-Cell-Interactions	264		
Genes in Embryonic Development	290		

Table of Contents in Detail

Introduction	1	DNA Repair	80
Chronology	13	Xeroderma Pigmentosum	82
Advances that Contributed to the Development of Genetics	13	Prokaryotic Cells and Viruses	84
Part 1. Fundamentals	19	Prokaryotic Cells	84
Molecular Basis of Genetics	20	Isolation of Mutant Bacteria	84
The Cell and Its Components	20	Recombination in Bacteria	86
Some Types of Chemical Bonds	22	Bacteriophages	88
Carbohydrates	24	DNA Transfer between Cells	90
Lipids (Fats)	26	Viruses	92
Nucleotides and Nucleic Acids	28	Replication Cycle of Viruses	94
Amino Acids	30	RNA Viruses: Genome, Replication, Translation	96
Proteins	32	DNA Viruses	98
DNA as Carrier of Genetic Information ..	34	Retroviruses	100
DNA and Its Components	36	Retrovirus Integration and Transcription	102
DNA Structure	38	Eukaryotic Cells	104
Alternative DNA Structures	40	Yeast: Eukaryotic Cells with a Diploid and a Haploid Phase	104
DNA Replication	42	Mating Type Determination in Yeast Cells and Yeast Two-Hybrid System	106
Genes	44	Functional Elements in Yeast Chromosomes	108
The Flow of Genetic Information: Transcription and Translation	44	Artificial Chromosomes for Analyzing Complex Genomes	110
Genes and Mutation	46	Cell Cycle Control	112
Genetic Code	48	Cell Division: Mitosis	114
The Structure of Eukaryotic Genes ...	50	Maturation Division (Meiosis)	116
Recombinant DNA	52	Crossing-Over in Prophase I	118
DNA Sequencing	52	Formation of Gametes	120
Automated DNA Sequencing	54	Cell Culture	122
DNA Cloning	56	Mitochondrial Genetics	124
cDNA Cloning	58	Genetically Controlled Energy-Delivering Processes in Mitochondria	124
DNA Libraries	60	The Genome in Chloroplasts and Mitochondria	126
Restriction Analysis by Southern Blot Analysis	62	The Mitochondrial Genome of Man	128
Restriction Mapping	64	Mitochondrial Diseases	130
DNA Amplification by Polymerase Chain Reaction (PCR)	66	Formal Genetics	132
Changes in DNA	68	The Mendelian Traits	132
Mutation due to Base Modifications ..	70	Segregation of Mendelian Traits	134
DNA Polymorphism	72		
Recombination	74		
Transposition	76		
Trinucleotide Repeat Expansion	78		

Independent Distribution of Two Different Traits	136	Control of Transcription	212
Phenotype and Genotype	138	Transcription Control in Eukaryotes	214
Segregation of Parental Genotypes	140	Regulation of Gene Expression in Eukaryotes	216
Monogenic Inheritance	142	DNA-Binding Proteins	218
Linkage and Recombination	144	Other Transcription Activators	220
Genetic Distance between Two Gene Loci ..	146	Inhibitors of Transcription and Translation	222
Analysis with Genetic Markers	148	DNA Methylation	224
Linkage Analysis	150	Genomic Imprinting	226
Quantitative Genetic Traits	152	X-Chromosome Inactivation	228
Normal Distribution and Polygenic Threshold Model	154	Targeted Gene Disruption in Transgenic Mice	230
Distribution of Genes in a Population ..	156		
Hardy-Weinberg Equilibrium	158	Part II. Genomics	233
Consanguinity and Inbreeding	160	Genomics, the Study of the Organization of Genomes	234
Twins	162	The Complete Sequence of the <i>Escherichia coli</i> Genome	236
Polymorphism	164	Genome of a Plasmid from a Multiresistant <i>Corynebacterium</i>	238
Biochemical Polymorphism	166	Genome Maps	240
Geographical Distribution of Genes	168	Approach to Genome Analysis	242
Chromosomes	170	Organization of Eukaryotic Genomes	244
Nucleosomes	170	Gene Identification	246
DNA in Chromosomes	172	The Human Genome Project	248
Polytene Chromosomes	174	Identification of a Coding DNA Segment ..	250
DNA in Lampbrush Chromosomes	176	The Dynamic Genome:	
Correlation of Structure and Function in Chromosomes	178	Mobile Genetic Elements	252
Special Structure at the Ends of a Chromosome: the Telomere	180	Evolution of Genes and Genomes	254
Metaphase Chromosomes	182	Comparative Genomics	256
Karyotype	184	Human Evolution	258
The G- and R-Banding Patterns of the Human Metaphase Chromosomes	186	Genome Analysis by DNA Microarrays ..	260
Designation of Chromosomal Aberrations	188		
Preparation of Metaphase Chromosomes ..	190	Part III. Genetics and Medicine	263
In Situ Hybridization	192	Cell-to-Cell Interactions	264
Specific Metaphase Chromosome Identification	194	Intracellular Signal Transduction Systems	264
Numerical Chromosome Aberrations	196	Types of Cell Surface Receptors	266
Translocation	198	G Protein-coupled Receptors	268
Structural Chromosomal Aberrations ..	200	Transmembrane Signal Transmitters	270
Detection of Structural Chromosomal Aberrations by Molecular Methods	202	Receptors of Neurotransmitters	272
Regulation and Expression of Genes	204	Genetic Defects in Ion Channels	274
The Cell Nucleus and Ribosomal RNA ..	204	Chloride Channel Defects:	
Transcription	206	Cystic Fibrosis	276
Control of Gene Expression in Bacteria by Induction	208	Rhodopsin, a Photoreceptor	278
Control of Gene Expression in Bacteria by Repression	210	Mutations in Rhodopsin	280
		Color Vision	282
		Hearing and Deafness	284

Odorant Receptor Gene Family	286	Lysosomes and LDL Receptor	352
Mammalian Taste Receptor Genes	288	Lysosomes and Endocytosis	352
Genes in Embryonic Development	290	Diseases Due to Lysosomal Enzyme Defects	354
Developmental Mutants in <i>Drosophila</i> ..	290	Mucopolysaccharide Storage Diseases ...	356
Homeobox Genes	292	Familial Hypercholesterolemia	358
Genetics in a Lucent Vertebrate Embryo: Zebrafish	294	Mutations in the LDL Receptor	360
Developmental Program for Individual Cells in the Nematode <i>C. elegans</i>	296	Homeostasis	362
Developmental Genes in a Plant Embryo (<i>Arabidopsis thaliana</i>)	298	Insulin and Diabetes Mellitus	362
Immune System	300	Protease Inhibitor α_1 -Antitrypsin	364
Components of the Immune System	300	Blood Coagulation Factor VIII (Hemophilia A)	366
Immunoglobulin Molecules	302	Von Willebrand Factors	368
Genetic Diversity Generated by Somatic Recombination	304	Cytochrome P450 Genes	370
Mechanisms in Immunoglobulin Gene Rearrangement	306	Pharmacogenetics	372
Genes of the MHC Region	308	Maintaining Cell and Tissue Shape	374
T-Cell Receptors	310	Cytoskeletal Proteins in Erythrocytes ...	374
Evolution of the Immunoglobulin Supergene Family	312	Hereditary Muscle Diseases	376
Hereditary and Acquired Immune Deficiencies	314	Duchenne Muscular Dystrophy	378
Origin of Tumors	316	Collagen Molecules	380
Influence of Growth Factors on Cell Division	316	Osteogenesis Imperfecta	382
Tumor Suppressor Genes	318	Molecular Basis of Bone Development ...	384
Cellular Oncogenes	320	Mammalian Sex Determination and Differentiation	386
The p53 Protein, a Guardian of the Genome	322	Sex Determination	386
Neurofibromatosis 1 and 2	324	Sex Differentiation	388
APC Gene in Familial Polyposis Coli	326	Disorders of Sexual Development	390
Breast Cancer Susceptibility Genes	328	Congenital Adrenal Hyperplasia	392
Retinoblastoma	330	Atypical Inheritance Pattern	394
Fusion Gene as Cause of Tumors: CML ...	332	Unstable Number of Trinucleotide Repeats	394
Genomic Instability Syndromes	334	Fragile X Syndrome	396
Oxygen and Electron Transport ...	336	Imprinting Diseases	398
Hemoglobin	336	Karyotype–Phenotype Correlation	400
Hemoglobin Genes	338	Autosomal Trisomies	400
Sickle Cell Anemia	340	Other Numerical Chromosomal Deviations	402
Mutations in Globin Genes	342	Deletions and Duplications	404
The Thalassemias	344	A Brief Guide to Genetic Diagnosis	406
Hereditary Persistence of Fetal Hemoglobin (HPFH)	346	Principles	406
DNA Analysis in Hemoglobin Disorders .	348	Detection of Mutations without Sequencing	408
Peroxisomal Diseases	350		

Chromosomal Location of Monogenic Diseases	410
General References	421
Glossary	423
Index	442

Introduction

Each of the approximately 10^{14} cells of an adult human contains a program with life-sustaining information in its nucleus. This allows an individual to interact with the environment not only through the sensory organs by being able to see, to hear, to taste, to feel heat, cold, and pain, and to communicate, but also to remember and to integrate the input into cognate behavior. It allows the conversion of atmospheric oxygen and ingested food into energy production and regulates the synthesis and transport of biologically important molecules. The immune defense against unwarranted invaders (e.g., viruses, bacteria, fungi) is part of the program. The shape and mobility of bones, muscles, and skin could not be maintained without it. The fate of each cell is determined by the control of cell division and differentiation into different types of cells and tissues, including cell-to-cell interactions and intracellular and extracellular signal transduction. Finally, such different areas as reproduction or the detoxification and excretion of molecules that are not needed depend on this program as well as many other functions of life.

This cellular program is genetically determined. It is transferred from one cell to both daughter cells at each cell division and from one generation to the next through specialized cells, the germ cells (oocytes and spermatozoa). The integrity of the genetic program must be maintained without compromise, yet it should also be adaptable to long-term changes in the environment. This is an enormous task. It is no wonder, therefore, that errors in maintaining and propagating the genetic program occur frequently in all living systems despite the existence of complex systems for damage recognition and repair.

All these biological processes are the result of biochemical reactions performed by biomolecules called proteins. Proteins are involved in the production of almost all molecules (including other proteins) in living cells. Proteins are made up of dozens to several hundreds of amino acids linearly connected to each other as a polypeptide, subsequently to be arranged in a specific three-dimensional structure, often in combination with other polypeptides. Only this latter feature allows biological function. Genetic information is the cell's blueprint to make the proteins that a specific cell typically makes. Most cells do not produce all possible proteins, but a selection depending on the type of cell.

Each of the 20 amino acids used by living organisms has a code of three specific chemical structures, the nucleotide bases, that are part of a large molecule, DNA (deoxyribonucleic acid). DNA is a read-only memory of the genetic information system. In contrast to the binary system of strings of ones and zeros used in computers ("bits", which are then combined into "bytes" that are eight binary digits long), the genetic code in the living world uses a quaternary system of four nucleotide bases with chemical names having the initial letters A, C, G, and T (see Part I, Fundamentals). With a quaternary code used in living cells the bytes (called "quytes" by The Economist in a Survey of the Human Genome, July 1, 2000) are shorter: three only, each called a triplet codon. Each linear sequence of amino acids in a protein is encoded by a corresponding sequence of codons in DNA (genetic code). The genetic code is universal and is used by all living cells, including plants and also by viruses. Each unit of genetic information is called a gene. This is the equivalent of a single sentence in a text. In fact, genetic information is highly analogous to a text and is amenable to being stored in computers.

Depending on the organizational complexity of the organism, the number of genes may be small as in viruses and bacteria (10 genes in a small bacteriophage or 4289 genes in *Escherichia coli*), medium (6241 genes in yeast; 13601 in *Drosophila*, 18 424 in a nematode), or large (about 80 000 in humans and other mammals). Since many proteins are involved in related functions of the same pathway, they and their corresponding genes can be grouped into families of related function. It is estimated that the human genes form about 1000 gene families. Each gene family arose by evolution from one ancestral gene or from a few. The entirety of genes and DNA in each cell of an organism is called the genome. By analogy, the entirety of proteins of an organism is called the proteome. The corresponding fields of study are termed genomics and proteomics, respectively.

Genes are located in chromosomes. These are individual, paired bodies consisting of DNA and special proteins in the cell nucleus. One chromosome of each homologous pair is derived from the mother and the other from the father. Man has 23 pairs. While the number and size of chromosomes in different organisms vary, the total amount of DNA and the total number of

genes are the same for a particular class of organism. Genes are arranged linearly along each chromosome. Each gene has a defined position (gene locus) and an individual structure and function. As a rule, genes in higher organisms are structured into contiguous sections of coding and noncoding sequences called exons (coding) and introns (noncoding), respectively. Genes in multicellular organisms vary with respect to overall size (a few thousand to over a million base pairs), number and size of exons, and regulatory DNA sequences that determine their state of activity, called the expression (most genes in differentiated, specialized cells are permanently turned off). It is remarkable that more than 90% of the total of 3 billion (3×10^9) base pairs of DNA in higher organisms do not carry any coding information (see Part II, Genomics).

The linear text of information contained in the coding sequences of DNA in a gene cannot be read directly. Rather, its total sequence is first transcribed into a structurally related molecule with a corresponding sequence of codons. This molecule is called RNA (ribonucleic acid) because it contains ribose instead of the deoxyribose of DNA. From this molecule the introns (from the noncoding parts) are then removed by special enzymes, and the exons (the coding parts) are spliced together into the final template, called messenger RNA (mRNA). From this molecule the corresponding encoded sequence of amino acids (polypeptide) is read off in a complex cellular machinery (ribosomes) in a process called translation.

Genes with the same, a similar, or a related function in different organisms are the same, similar, or related in certain ways. This is expressed as the degree of structural or functional similarity. The reason for this is evolution. All living organisms are related to each other because their genes are related. In the living world, specialized functions have evolved but once, encoded by the corresponding genes. Therefore, the structures of genes required for fundamental functions are preserved across a wide variety of organisms, for example functions in cell cycle control or in embryonic development and differentiation. Such genes are similar or identical even in organisms quite distantly related, such as yeast, insects, worms, vertebrates, mammals, and even plants. Such genes of fundamental importance do not

tolerate changes (mutations), because this would compromise function. As a result, deleterious mutations do not accumulate in any substantial number. Similar or identical genes present in different organisms are referred to as conserved in evolution. All living organisms have elaborate cellular systems that can recognize and eliminate faults in the integrity of DNA and genes (DNA repair). Mechanisms exist to sacrifice a cell by programmed cell death (apoptosis) if the defect cannot be successfully repaired.

Unlike the important structures that time has evolutionarily conserved, DNA sequences of no or of limited direct individual importance differ even among individuals of the same species. These individual differences (genetic polymorphism) constitute the genetic basis for the uniqueness of each individual. At least one in 1000 base pairs of human DNA differs among individuals (single nucleotide polymorphism, SNP). In addition, many other forms of DNA polymorphism exist that reflect a high degree of individual genetic diversity.

Individual genetic differences in the efficiency of metabolic pathways are thought to predispose to diseases that result from the interaction of many genes, often in combination with particular environmental influences. They may also protect one individual from an illness to which another is prone. Such individual genetic differences are targets of individual therapies by specifically designed pharmaceutical substances aimed at high efficacy and low risk of side effects (pharmacogenomics). The Human Genome Project should greatly contribute to the development of an individual approach to diagnostics and therapy (genetic medicine).

Human populations of different geographic origins also are related by evolution (see section on human evolution in Part II). They are often mistakenly referred to as races. Modern mankind originated in Africa about 200 000 years ago and had migrated to all parts of the world by about 100 000 years ago. Owing to regional adaptation to climatic and other conditions, and favored by geographic isolation, different ethnic groups evolved. They are recognizable by literally superficial features, such as color of the skin, eyes, and hair, that betray the low degree of human genetic variation between different populations. Genetically speaking, *Homo sapiens* is one rather homogeneous species of re-

cent origin. Of the total genetic variation, about 85% is interindividual within a given group, only 15% is among different groups (populations). In contrast, chimpanzees from one group in West Africa are genetically more diverse than all humans ever studied. As a result of evolutionary history, humans are well adapted to live peacefully in relatively small groups with a similar cultural and linguistic background. Unfortunately, humans are not yet adapted to global conditions. They tend to react with hostility to groups with a different cultural background in spite of negligible genetic differences. Genetics does not provide any scientific basis for claims that favor discrimination, but it does provide direct evidence for the evolution of life on earth. Genetics is the science concerned with the structure and function of all genes in different organisms (analysis of biological variation). New investigative methods and observations, especially during the last 10 to 20 years, have helped to integrate this field into the mainstream of biology and medicine. Today, it plays a central, unifying role comparable to that of cellular pathology at the beginning of the last century. Genetics is relevant to virtually all medical specialties. Knowledge of basic genetic principles and their application in diagnosis are becoming an essential part of medical education today.

Classical Genetics Between 1900 and 1953

(see *chronological table on p. 13*)

In 1906, the English biologist William Bateson (1861 – 1926) proposed the term *genetics* for the new biological field devoted to investigating the rules governing heredity and variation. Bateson referred to heredity and variation when comparing the similarities and differences, respectively, of genealogically related organisms, two aspects of the same phenomenon. Bateson clearly recognized the significance of the Mendelian rules, which had been rediscovered in 1900 by Correns, Tschermak, and DeVries.

The Mendelian rules are named for the Augustinian monk Gregor Mendel (1822 – 1884), who conducted crossbreeding experiments on garden peas in his monastery garden in Brünn (Brno, Czech Republic) well over a century ago. In 1866, Mendel wrote that heredity is based on individual factors that are indepen-



Johann Gregor Mendel

dent of each other (see Brink and Styles, 1965; Mayr, 1982). Transmission of these factors to the next plant generation, i.e., the distribution of different traits among the offspring, occurred in predictable proportions. Each factor was responsible for a certain trait. The term *gene* for such a heritable factor was introduced in 1909 by the Danish biologist Wilhelm Johannsen (1857 – 1927).

Starting in 1902, Mendelian inheritance was systematically analyzed in animals, plants, and also in man. Some human diseases were recognized as having a hereditary cause. A form of brachydactyly (type A1, McKusick 112500) observed in a large Pennsylvania sibship by W. C. Farabee (PhD thesis, Harvard University, 1902) was the first condition in man to be described as being transmitted by autosomal dominant inheritance (Haws and McKusick, 1963).

In 1909, Archibald Garrod (1857 – 1936), later Regius Professor of Medicine at Oxford University, demonstrated that four congenital metabolic diseases (albinism, alkaptonuria, cystinuria, and pentosuria) are transmitted by autosomal recessive inheritance (Garrod, 1909). Garrod was the first to recognize that there are biochemical differences among individuals that do not lead to illness but that have a genetic

basis. However, the relationship of genetic and biochemical findings revealed by this concept was ahead of its time: the far-reaching significance for the genetic individuality of man was not recognized (Bearn, 1993). Certainly part of the reason was that the nature of genes and how they function was completely unclear. Early genetics was not based on chemistry or on cytology (Dunn, 1965; Sturtevant, 1965). Chromosomes in mitosis (Flemming, 1879) and meiosis (Strasburger, 1888) were observed; the term chromosome was coined by Waldeyer in 1888, but a functional relationship between genes and chromosomes was not considered. An exception was the prescient work of Theodor Boveri (1862–1915) about the genetic individuality of chromosomes (in 1902).

Genetics became an independent scientific field in 1910 with the study of the fruit fly (*Drosophila melanogaster*) by Thomas H. Morgan at Columbia University in New York. Subsequent systematic genetic studies on *Drosophila* over many years (Dunn, 1965; Sturtevant, 1965; Whitehouse, 1973) showed that genes are arranged linearly on chromosomes. This led to the chromosome theory of inheritance (Morgan, 1915).



Thomas H. Morgan

The English mathematician Hardy and the German physician Weinberg recognized that Mendelian inheritance accounts for certain regularities in the genetic structure of populations (1908). Their work contributed to the successful introduction of genetic concepts into plant and animal breeding. Although genetics was well established as a biological field by the end of the third decade of last century, knowledge of the physical and chemical nature of genes was sorely lacking. Structure and function remained unknown.

That genes can change and become altered was recognized by DeVries in 1901. He introduced the term mutation. In 1927, H. J. Muller determined the spontaneous mutation rate in *Drosophila* and demonstrated that mutations can be induced by roentgen rays. C. Auerbach and J. M. Robson (1941) and, independently, F. Oehlkers (1943) observed that certain chemical substances also could induce mutations. However, it remained unclear what a mutation actually was, since the physical basis for the transfer of genetic information was not known.

The complete lack of knowledge of the structure and function of genes contributed to misconceptions in the 1920s and 30s about the possibility of eliminating “bad genes” from human populations (eugenics). However, modern genetics has shown that the ill conceived eugenic approach to eliminating human genetic disease is also ineffective.

Thus, incomplete genetic knowledge was applied to human individuals at a time when nothing was known about the structure of genes. Indeed, up to 1949 no essential genetic findings had been gained from studies in man. Quite the opposite holds true today.

Today, it is evident that genetically determined diseases generally cannot be eradicated. No one is free from a genetic burden. Every individual carries about five or six severe genetic defects that are inapparent, but that may show up in offspring.

With the demonstration in the fungus *Neurospora* that one gene is responsible for the formation of one enzyme (“one gene, one enzyme”, Beadle and Tatum, 1941), the close relationship of genetics and biochemistry became apparent, quite in agreement with Garrod’s concept of in-born errors of metabolism. Systematic studies in microorganisms led to other important advances in the 1940s: genetic recombination was

demonstrated in bacteria (Lederberg and Tatum, 1946) and viruses (Delbrück and Bailey, 1947). Spontaneous mutations were observed in bacterial viruses (bacteriophages; Hershey, 1947). The study of genetic phenomena in microorganisms turned out to be as significant for the further development of genetics as the analysis of *Drosophila* had been 35 years earlier (for review, see Cairns et al., 1978). A very influential, small book entitled "What Is Life?" by the physicist E. Schrödinger (1944) defined genes in molecular terms. At that time, elucidation of the molecular biology of the gene became a central theme in genetics.

Genetics and DNA

A major advance occurred in 1944 when Avery, MacLeod, and McCarty at the Rockefeller Institute in New York demonstrated that a chemically relatively simple long-chained nucleic acid (deoxyribonucleic acid, DNA) carried genetic information in bacteria (for historical review, see Dubos, 1976; McCarty, 1985). Many years earlier, F. Griffith (in 1928) had observed that permanent (genetic) changes can be induced in pneumococcal bacteria by a cell-free extract derived from other strains of pneumococci ("transforming principle"). Avery and his co-workers showed that DNA was this transforming principle. In 1952, Hershey and Chase proved that genetic information is transferred by DNA alone. With this knowledge, the question of its structure became paramount.

This was resolved most elegantly by James D. Watson, a 24-year-old American on a scholarship in Europe, and Francis H. Crick, a 36-year-old English physicist, at the Cavendish Laboratory of the University of Cambridge. Their findings appeared in a three-quarter-page article on April 25, 1953 in *Nature* (Watson and Crick, 1953). In this famous article, Watson and Crick proposed that the structure of DNA is a double helix. The double helix is formed by two complementary chains with oppositely oriented alternating sugar (deoxyribose) and monophosphate molecules. Inside this helical molecule lie paired nucleotide bases, each pair consisting of a purine and a pyrimidine. The crucial feature is that the base pairs lie inside the molecule, not outside. This insight came from construction of a model of DNA that took into account stereochemical considerations and the results of previous X-ray diffraction studies



Oswald T. Avery

by M. Wilkins and R. Franklin. That the authors fully recognized the significance for genetics of the novel structure is apparent from the closing statement of their article, in which they state, "It has not escaped our notice that the specific pairing we have postulated immediately suggests a possible copying mechanism for the genetic material." Vivid, albeit different, accounts of their discovery have been given by the authors (Watson, 1968; Crick, 1988).

The elucidation of the structure of DNA is regarded as the beginning of a new era of molecular biology and genetics. The description of DNA as a double-helix structure led directly to an understanding of the possible structure of genetic information.

When F. Sanger determined the sequence of amino acids of insulin in 1955, he provided the first proof of the primary structure of a protein. This supported the notion that the sequence of amino acids in proteins could correspond with the sequential character of DNA. However, since DNA is located in the cell nucleus and protein synthesis occurs in the cytoplasm, DNA could not act directly. It turned out that DNA is first transcribed into a chemically similar mes-

senger molecule (messenger ribonucleic acid, mRNA) (Crick, Barnett, Brenner, Watts-Tobin 1961) with a corresponding nucleotide sequence, which is transported into the cytoplasm. In the cytoplasm, the mRNA then serves as a template for the amino acid sequence to be formed. The genetic code for the synthesis of proteins from DNA and messenger RNA was determined in the years 1963–1966 (Nirenberg, Mathaei, Ochoa, Benzer, Khorana, and others). Detailed accounts of these developments have been presented by Chargaff (1978), Judson (1996), Stent (1981), Watson and Tooze (1981), Crick (1988), and others.

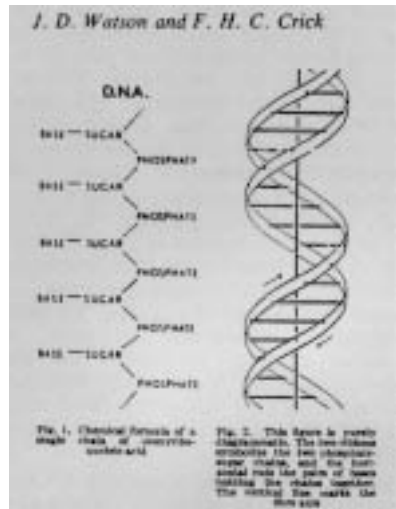
Important Methodological Advances in the Development of Genetics after About 1950

From the beginning, genetics has been a field strongly influenced by the development of new experimental methods. In the 1950s and 1960s, the groundwork was laid for biochemical genetics and immunogenetics. Relatively simple but reliable procedures for separating complex molecules by different forms of electrophoresis, methods for synthesizing DNA in vitro (Kornberg, 1956), and other approaches were applied to questions in genetics. The development of cell culture methods was of particular importance for the genetic analysis of humans. Pontecorvo introduced the genetic analysis of cultured eukaryotic cells (somatic cell genetics) in 1958. The study of mammalian genetics, with increasing significance for studying human genes, was facilitated by methods for fusing cells in culture (cell hybridization, T. Puck, G. Barski, B. Ephrussi, 1961) and the development of a cell culture medium for selecting certain mutants in cultured cells (HAT medium, J. Littlefield, 1964). The genetic approach that had been so successful in bacteria and viruses could now be applied in higher organisms, thus avoiding the obstacles of a long generation time and breeding experiments. A hereditary metabolic defect of man (galactosemia) was demonstrated for the first time in cultured human cells in 1961 (Krooth). The correct number of chromosomes in man was determined in 1956 (Tjio and Levan; Ford and Hamerton). Lymphocyte cultures were introduced for chromosomal analysis (Hungerford et al., 1960). The replication pattern of human chromosomes was described (J. German, 1962). These developments

further paved the way for expansion of the new field of human genetics.

Human Genetics

The medical aspects of human genetics (medical genetics) came to attention when it was recognized that sickle cell anemia is hereditary (Neel, 1949) and caused by a defined alteration of normal hemoglobin (Pauling, Itano, Singer, and Wells 1949), and again when it was shown that an enzyme defect (glucose-6-phosphatase



DNA structure 1953



Watson (left) and Crick (right) in 1953

deficiency, demonstrated in liver tissue by Cori and Cori in 1952) was the cause of a hereditary metabolic disease in man (glycogen storage disease type I, or von Gierke disease). The American Society of Human Genetics and the first journal of human genetics (*American Journal of Human Genetics*) were established in 1949. In addition, the first textbook of human genetics appeared (Curt Stern, *Principles of Human Genetics*, 1949).

In 1959, chromosomal aberrations were discovered in some well-known human disorders (trisomy 21 in Down syndrome by J. Lejeune, M. Gautier, R. Turpin; 45,X0 in Turner syndrome by Ford et al.; 47,XXY in Klinefelter syndrome by Jacobs and Strong). Subsequently, other numerical chromosome aberrations were shown to cause recognizable diseases in man (trisomy 13 and trisomy 18, by Patau et al. and Edwards et al. in 1960, respectively), and loss of small parts of chromosomes were shown to be associated with recognizable patterns of severe developmental defects (Lejeune et al., 1963; Wolf, 1964; Hirschhorn, 1964). The Philadelphia chromosome, a characteristic structural alteration of a chromosome in bone marrow cells of patients with adult type chronic myelogenous leukemia, was described by Nowell and Hungerford in 1962. The central role of the Y chromosome in establishing gender in mammals became apparent when it was realized that individuals without a Y chromosome are female and individuals with a Y chromosome are male, irrespective of the number of X chromosomes present. These observations further promoted interest in a new subspecialty, human cytogenetics.

Since early 1960, important knowledge about genetics in general has been obtained, often for the first time, by studies in man. Analysis of genetically determined diseases in man has yielded important insights into the normal function of genes in other organisms as well. Today, more is known about the general genetics of man than about that of any other species. Numerous subspecialties of human genetics have arisen, such as biochemical genetics, immunogenetics, somatic cell genetics, cytogenetics, clinical genetics, population genetics, teratology, mutational studies, and others. The development of the field has been well summarized by Vogel and Motulsky (1997) and McKusick (1992).

Genetics and Medicine

Most disease processes can be viewed as resulting from environmental influences interacting with the individual genetic makeup of the affected individual. A disease is genetically determined if it is mainly or exclusively caused by disorders in the genetic program of cells and tissues. More than 3000 defined human genetic diseases are known to be due to a mutation at a single gene locus (monogenic disease) and to follow a Mendelian mode of inheritance (McKusick 1998). They differ as much as the genetic information in the genes involved and may be manifest in essentially all age groups and organ systems. An important category of disease results from genetic predisposition interacting with precipitating environmental factors (multigenic or multifactorial diseases). This includes many relatively common chronic diseases (e.g., high blood pressure, hyperlipidemia, diabetes mellitus, gout, psychiatric disorders, certain congenital malformations). Further categories of genetically determined diseases are nonhereditary disorders in somatic cells (different forms of cancer) and chromosomal aberrations.

Due to new mutations and small family size in developed countries, genetic disorders usually do not affect more than one member of a family. About 90% occur as isolated cases within a family. Thus, their genetic origin cannot be recognized by familial aggregation. Instead, they must be recognized by their clinical features. This may be difficult in view of the many different functions of genes in normal tissues and in disease. Since genetic disorders affect all organ systems and age groups and are frequently not recognized, their contribution to the causes of human diseases appears smaller than it actually is. Genetically determined diseases are not a marginal group, but make up a substantial proportion of diseases. More than one-third of all pediatric hospital admissions are for diseases and developmental disorders that, at least in part, are caused by genetic factors (Weatherall 1991). The total estimated frequency of genetically determined diseases of different categories in the general population is about 3.5–5.0% (see Table 1).

The large number of individually rare genetically determined diseases and the overlap of diseases with similar clinical manifestations

Table 1 Frequency of genetically determined diseases

Type of genetic disease	Frequency per 1000 individuals
1. Monogenic diseases, total	4.5 – 15.0
Autosomal dominant	2 – 9.5
Autosomal recessive	2 – 3.5
X-chromosomal	0.5 – 2
2. Chromosomal aberrations	5 – 7
3. Multifactorial disorders*	70 – 90
4. Congenital malformations	19 – 22
Total	ca. 80 – 115

* Contribution of genetic factors variable. (Data based on Weatherall, 1991.)

but different etiology (principle of genetic or etiological heterogeneity) cause additional diagnostic difficulties. This must be considered during diagnosis to avoid false conclusions about a genetic risk.

In 1966 Victor A. McKusick introduced a catalog of human phenotypes transmitted according to Mendelian inheritance (McKusick catalog, currently in its 12th edition; McKusick 1998). This catalog and the 1968 – 1973 Baltimore Conferences organized by McKusick (Clinical Delineation of Birth Defects) have contributed substantially to the systematization and subsequent development of medical genetics. The extent of medical genetics is reflected by the initiation of several new scientific journals since 1965 (*Clinical Genetics*, *Journal of Medical Genetics*, *Human Genetics*, *Annales de Génétique*, *American Journal of Medical Genetics*, *Cytogenetics and Cell Genetics*, *European Journal of Human Genetics*, *Prenatal Diagnosis*, *Clinical Dysmorphology*, and others).

In recent years, considerable, previously unexpected progress in clarifying the genetic etiology of human diseases, and thereby in furnishing insights into the structure and function of normal genes, has been achieved by molecular methods.

Molecular Genetics

The discovery in 1970 (independently by H. Temin and D. Baltimore) of reverse transcriptase, an unusual enzyme complex in RNA viruses (retroviruses), upset the dogma—valid up to that time—that the flow of genetic infor-

mation went in one direction only, i.e., from DNA to RNA and from there to the gene product (a peptide). Not only is the existence of reverse transcriptase an important biological finding, but the enzyme provides a means of obtaining complementary DNA (cDNA) that corresponds to the coding regions of an active gene. Therefore, it is possible to analyze a gene directly without knowledge of its gene product, provided it is expressed in the tissue examined.

In addition, enzymes that cleave DNA at specific sites (restriction endonucleases or, simply, restriction enzymes) were discovered in bacteria (W. Arber, 1969; D. Nathans and H. O. Smith, 1971). With appropriate restriction enzymes, DNA can be cut into pieces of reproducible and defined size, thus allowing easy recognition of an area to be studied. DNA fragments of different origin can be joined and their properties analyzed. Methods for producing multiple copies of DNA fragments and sequencing them (determining the sequence of their nucleotide bases) were developed between 1977 and 1985. These methods are collectively referred to as recombinant DNA technology (see Chronology at the end of this introduction).

In 1977, recombinant DNA analysis led to a completely new and unexpected finding about the structure of genes in higher organisms, but also in yeast and *Drosophila*: Genes are not continuous segments of coding DNA, but are usually interrupted by noncoding segments (see Watson and Tooze 1981; Watson et al., 1992). The size and sequence of coding DNA segments, or exons (a term introduced by Gilbert in 1978), and noncoding segments, or introns, are

specific for each individual gene (exon/intron structure of eukaryotic genes).

With the advent of molecular genetic DNA analysis, many different types of polymorphic DNA markers, i.e., individual heritable differences in the nucleotide sequence, have been mapped to specific sites on chromosomes (physical map). As a result, the chromosomal position of a gene of interest can now be determined (mapped) by analyzing the segregation of a disease locus in relation to the polymorphic DNA markers (linkage analysis). Once the chromosomal location of a gene is known, the latter can be isolated and its structure can be characterized (positional cloning, a term introduced by F. Collins). The advantage of such a direct analysis is that nothing needs to be known about the gene of interest aside from its approximate location. Prior knowledge of the gene product is not required.

Another, complementary, approach is to identify a gene with possible functional relevance to a disorder (a candidate gene), determine its chromosomal position, and then demonstrate mutations in the candidate gene in patients with the disorder. Positional cloning and identification of candidate genes have helped identify genes for many important diseases such as achondroplasia, degenerative retinal diseases, cystic fibrosis, Huntington chorea and other neurodegenerative diseases, Duchenne muscular dystrophy and other muscular diseases, mesenchymal diseases with collagen defects (osteogenesis imperfecta), Marfan syndrome (due to a defect of a previously unknown protein, fibrillin), immune defects, and numerous tumors.

The extensive homologies of genes that regulate embryological development in different organisms and the similarities of genome structures have contributed to leveling the boundaries in genetic analysis that formerly existed for different organisms (e.g., *Drosophila* genetics, mammalian genetics, yeast genetics, bacterial genetics, etc.). Genetics has become a broad, unifying discipline in biology, medicine, and evolutionary research.

The Dynamic Genome

Between 1950 und 1953, remarkable papers appeared entitled "The origin and behavior of mutable loci in maize" (*Proc Natl Acad Sci.* 36: 344–355, 1950), "Chromosome organization and

genic expression" (*Cold Spring Harbor Symp Quant Biol.* 16: 13–45, 1952), and "Introduction of instability at selected loci in maize" (*Genetics* 38: 579–599, 1953). Here the author, Barbara McClintock of Cold Spring Harbor Laboratory, described mutable loci in Indian corn plants (maize) and their effect on the phenotype of corn due to a gene that is not located at the site of the mutation. Surprisingly, this gene can exert a type of remote control. In addition, other genes can change their location and cause mutations at distant sites.

In subsequent work, McClintock described the special properties of this group of genes, which she called controlling genetic elements (*Brookhaven Symp Biol.* 8: 58–74, 1955). Different controlling elements could be distinguished according to their effects on other genes and the mutations caused. However, her work received little interest (for review see Fox Keller 1983; Fedoroff and Botstein 1992).

Thirty years later, at her 1983 Nobel Prize lecture ("The significance of responses of the genome to challenge," *Science* 226: 792–801, 1984), things had changed. Today we know that the genome is not rigid and static. Rather, it is flexible and dynamic because it contains parts that can move from one location to another (mobile genetic elements, the current designation). The precision of the genetic information depends on its stability, but complete stability would also mean static persistence. This would be detrimental to the development of new forms of life in response to environmental changes. Thus, the genome is subject to alterations, as life requires a balance between the old and the new.

The Human Genome Project

A new dimension has been introduced into biomedical research by the Human Genome Project (HGP) and related programs in many other organisms (see Part II, Genomics). The main goal of the HGP is to determine the entire sequence of the 3 billion nucleotide pairs in the DNA of the human genome and to find all the genes within it. This is a daunting task. It is comparable to deciphering each individual 1-mm-wide letter along a 3000-km-long text strip. As more than 90% of DNA is not part of genes, other approaches aimed at expressed (active) genes are taken. The completion of a draft covering about 90% of the genome was announced in

June 2000 (*Nature* June 29, 2000, pp. 983–985; *Science* June 30, pp. 2304–2307). The complete sequence of human chromosomes 22 and 21 was published in late 1999 and early 2000, respectively. Conceived in 1986 and officially begun in 1990, the HGP has progressed at a brisk pace. It is expected to be completed in 2003, several years ahead of the original plan (for a review see Lander and Weinberg, 2000, and Part II, Genomics).

Ethical and Societal Aspects

From its start the Human Genome Project devoted attention and resources to ethical, legal, and social issues (the ELSI program). This is an important part of the HGP in view of the far-reaching consequences of the current and expected knowledge about human genes and the genome. Here only a few areas can be mentioned. Among these are questions of validity and confidentiality of genetic data, of how to decide about a genetic test prior to the first manifestation of a disease (presymptomatic genetic testing), or whether to test for the presence or absence of a disease-causing mutation in an individual before any signs of the disease can be expected (predictive genetic testing). How does one determine whether a genetic test is in the best interest of the individual? Does she or he benefit from the information, could it result in discrimination? How are the consequences defined? How is (genetic) counseling done and informed consent obtained? The use of embryonic stem cells is another area that concerns the public. Careful consideration of benefits and risks in the public domain will aid in reaching rational and balanced decisions.

Education

Although genetic principles are rather straightforward, genetics is opposed by some and misunderstood by many. Scientists should seize any opportunity to inform the public about the goals of genetics and genomics and the principal methods employed. Genetics should be highly visible at the elementary and high school levels. Human genetics should be emphasized in teaching in medical schools.

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Chronology

Advances that Contributed to the Development of Genetics

(This list contains selected events and should not be considered complete, especially for the many important developments during the past several years.)

- | | | | |
|------|--|------|---|
| 1839 | Cells recognized as the basis of living organisms (<i>Schleiden, Schwann</i>) | 1911 | Sarcoma virus (<i>Peyton Rous</i>) |
| 1859 | Concepts of evolution (<i>Charles Darwin</i>) | 1912 | Crossing-over (<i>Morgan and Cattell</i>)
Genetic linkage (<i>Morgan and Lynch</i>)
First genetic map (<i>A. H. Sturtevant</i>) |
| 1865 | Rules of inheritance by distinct "factors" acting dominantly or recessively (<i>Gregor Mendel</i>) | 1913 | First cell culture (<i>A. Carrel</i>) |
| 1869 | "Nuclein," a new acidic, phosphorus-containing, long molecule (<i>F. Miescher</i>) | 1914 | Nondisjunction (<i>C. B. Bridges</i>) |
| 1879 | Chromosomes in mitosis (<i>W. Flemming</i>) | 1915 | Genes located on chromosomes (chromosomal theory of inheritance) (<i>Morgan, Sturtevant, Muller, Bridges</i>) |
| 1883 | Quantitative aspects of heredity (<i>F. Galton</i>) | 1922 | Characteristic phenotypes of different trisomies in the plant <i>Datura stramonium</i> (<i>F. Blakeslee</i>) |
| 1889 | Term "nucleic acid" introduced (<i>R. Altmann</i>) | 1924 | Blood group genetics (<i>Bernstein</i>)
Statistical analysis of genetic traits (<i>Fisher</i>) |
| 1892 | Term "virus" introduced (<i>R. Ivanowski</i>) | 1926 | Enzymes are proteins (<i>J. Sumner</i>) |
| 1897 | Enzymes discovered (<i>E. Büchner</i>) | 1927 | Mutations induced by X-rays (<i>H. J. Muller</i>)
Genetic drift (<i>S. Wright</i>) |
| 1900 | Mendel's discovery recognized (<i>H. de Vries, E. Tschermak, K. Correns</i> , independently)
ABO blood group system (<i>Landsteiner</i>) | 1928 | Euchromatin/heterochromatin (<i>E. Heitz</i>)
Genetic transformation in pneumococci (<i>F. Griffith</i>) |
| 1902 | Some diseases in man inherited according to Mendelian rules (<i>W. Bateson, A. Garrod</i>)
Individuality of chromosomes (<i>T. Boveri</i>)
Chromosomes and Mendel's factors are related (<i>W. Sutton</i>)
Sex chromosomes (<i>McClung</i>) | 1933 | Pedigree analysis (<i>Haldane, Hogben, Fisher, Lenz, Bernstein</i>)
Polytene chromosomes (<i>Heitz and Bauer, Painter</i>) |
| 1906 | Term "genetics" proposed (<i>W. Bateson</i>) | 1935 | First cytogenetic map in <i>Drosophila</i> (<i>C. B. Bridges</i>) |
| 1908 | Population genetics (<i>Hardy, Weinberg</i>) | 1937 | Mouse H2 gene locus (<i>P. Gorer</i>) |
| 1909 | Inborn errors of metabolism (<i>Garrod</i>)
Terms "gene," "genotype," "phenotype" proposed (<i>W. Johannsen</i>)
Chiasma formation during meiosis (<i>Janssens</i>)
First inbred mouse strain DBA (<i>C. Little</i>) | 1940 | Polymorphism (<i>E. B. Ford</i>)
Rhesus blood groups (<i>Landsteiner and Wiener</i>) |
| 1910 | Beginning of <i>Drosophila</i> genetics (<i>T. H. Morgan</i>)
First <i>Drosophila</i> mutation (<i>white-eyed</i>) | 1941 | Evolution through gene duplication (<i>E. B. Lewis</i>)
Genetic control of enzymatic biochemical reactions (<i>Beadle and Tatum</i>)
Mutations induced by mustard gas (<i>Auerbach</i>) |
| | | 1944 | DNA as the material basis of genetic information (<i>Avery, MacLeod, McCarty</i>)
"What is life? The Physical Aspect of the Living Cell." An influential book (<i>E. Schrödinger</i>) |
| | | 1946 | Genetic recombination in bacteria (<i>Lederberg and Tatum</i>) |

- 1947 Genetic recombination in viruses (*Delbrück and Bailey, Hershey*)
- 1949 Sickle cell anemia, a genetically determined molecular disease (*Neel, Pauling*)
Hemoglobin disorders prevalent in areas of malaria (*J. B. S. Haldane*)
X chromatin (*Barr and Bertram*)
- 1950 A defined relation of the four nucleotide bases (*Chargaff*)
- 1951 Mobile genetic elements in Indian corn (*Zea mays*) (*B. McClintock*)
- 1952 Genes consist of DNA (*Hershey and Chase*)
Plasmids (*Lederberg*)
Transduction by phages (*Zinder and Lederberg*)
First enzyme defect in man (*Cori and Cori*)
First linkage group in man (*Mohr*)
Colchicine and hypotonic treatment in chromosomal analysis (*Hsu and Pomerat*)
Exogenous factors as a cause of congenital malformations (*J. Warkany*)
- 1953 DNA structure (*Watson and Crick, Franklin, Wilkins*)
Nonmendelian inheritance (*Ephrussi*)
Cell cycle (*Howard and Pelc*)
Dietary treatment of phenylketonuria (*Bickel*)
- 1954 DNA repair (*Muller*)
Leukocyte drumsticks (*Davidson and Smith*)
Cells in Turner syndrome are X-chromatin negative (*Polani*)
- 1955 Amino acid sequence of insulin (*F. Sanger*)
Lysosomes (*C. de Duve*)
Buccal smear (*Moore, Barr, Marberger*)
5-Bromouracil, an analogue of thymine, induces mutations in phages (*A. Pardee and R. Litman*)
- 1956 46 Chromosomes in man (*Tijo and Levan, Ford and Hamerton*)
DNA synthesis in vitro (*Kornberg*)
Genetic heterogeneity (*Harris, Fraser*)
- 1957 Amino acid sequence of hemoglobin molecule (*Ingram*)
- Cistron, the smallest nonrecombinant unit of a gene (*Benzer*)
Genetic complementation (*Fincham*)
DNA replication is semiconservative (*Meselson and Stahl, Taylor, Delbrück, Stent*)
Genetic analysis of radiation effects in man (*Neel and Schull*)
- 1958 Somatic cell genetics (*Pontocorvo*)
Ribosomes (*Roberts, Dintzis*)
Human HLA antigens (*Dausset*)
Cloning of single cells (*Sanford, Puck*)
Synaptonemal complex, the area of synapse in meiosis (*Moses*)
- 1959 First chromosomal aberrations described in man: trisomy 21 (*Lejeune, Gautier, Turpin*), Turner syndrome: 45,XO (*Jacobs*), Klinefelter syndrome: 47,XXY (*Ford*)
Isoenzymes (*Vesell, Markert*)
Pharmacogenetics (*Motulsky, Vogel*)
- 1960 Phytohemagglutinin-stimulated lymphocyte cultures (*Nowell, Moorehead, Hungerford*)
- 1961 The genetic code is read in triplets (*Crick, Brenner, Barnett, Watts-Tobin*)
The genetic code determined (*Nirenberg, Mathaei, Ochoa*)
X-chromosome inactivation (*M. F. Lyon*, confirmed by *Beutler, Russell, Ohno*)
Gene regulation, concept of operon (*Jacob and Monod*)
Galactosemia in cell culture (*Krooth*)
Cell hybridization (*Barski, Ephrussi*)
Thalidomide embryopathy (*Lenz, McBride*)
- 1962 Philadelphia chromosome (*Nowell and Hungerford*)
Xg, the first X-linked human blood group (*Mann, Race, Sanger*)
Screening for phenylketonuria (*Guthrie, Bickel*)
Molecular characterization of immunoglobulins (*Edelman, Franklin*)
Identification of individual human chromosomes by ³H-autoradiography (*German, Miller*)
Replicon (*Jacob and Brenner*)
Term "codon" for a triplet of (sequential) bases (*S. Brenner*)

- 1963 Lysosomal storage diseases (*C. de Duve*)
First autosomal deletion syndrome (cri-du-chat syndrome) (*J. Lejeune*)
- 1964 Excision repair (*Setlow*)
MLC test (*Bach and Hirschhorn, Bain and Lowenstein*)
Microlymphotoxicity test (*Terasaki and McClelland*)
Selective cell culture medium HAT (*Littlefield*)
Spontaneous chromosomal instability (*German, Schroeder*)
Cell fusion with Sendai virus (*Harris and Watkins*)
Cell culture from amniotic fluid cells (*H. P. Klinger*)
Hereditary diseases studied in cell cultures (*Danes, Bearn, Krooth, Mellman*)
Population cytogenetics (*Court Brown*)
Fetal chromosomal aberrations in spontaneous abortions (*Carr, Benirschke*)
- 1965 Limited life span of cultured fibroblasts (*Hayflick, Moorehead*)
- 1966 Catalog of Mendelian phenotypes in man (*McKusick*)
- 1968 HLA-D the strongest histocompatibility system (*Ceppellini, Amos*)
Repetitive DNA (*Britten and Kohne*)
Biochemical basis of the ABO blood group substances (*Watkins*)
DNA excision repair defect in xeroderma pigmentosum (*Cleaver*)
Restriction endonucleases (*H. O. Smith, Linn and Arber; Meselson and Yuan*)
First assignment of an autosomal gene locus in man (*Donahue, McKusick*)
Synthesis of a gene in vitro (*Khorana*)
- 1970 Reverse transcriptase (*D. Baltimore, H. Temin*, independently)
Synteny, a new term to refer to all gene loci on the same chromosome (*Renwick*)
Enzyme defects in lysosomal storage diseases (*Neufeld, Dorfman*)
Individual chromosomal identification by specific banding stains (*Zech, Casper, Lubs, Drets and Shaw, Schnedl, Evans*)
Y-chromatin (*Pearson, Bobrow, Vosa*)
Thymus transplantation for immune deficiency (*van Bekkum*)
- 1971 Two-hit theory in retinoblastoma (*A. G. Knudson*)
- 1972 High average heterozygosity (*Harris and Hopkinson, Lewontin*)
Association of HLA antigens and diseases
- 1973 Receptor defects in the etiology of genetic defects, genetic hyperlipidemia (*Brown, Goldstein, Motulsky*)
Demonstration of sister chromatid exchanges with BrdU (*S. A. Latt*)
Philadelphia chromosome as translocation (*J. D. Rowley*)
- 1974 Chromatin structure, nucleosome (*Kornberg, Olins and Olins*)
Dual recognition of foreign antigen and HLA antigen by T lymphocytes (*P. C. Doherty and R. M. Zinkernagel*)
Clone of a eukaryotic DNA segment mapped to a specific chromosomal location (*D. S. Hogness*)
- 1975 Asilomar conference
First protein-signal sequence identified (*G. Blobel*)
Southern blot hybridization (*E. Southern*)
Monoclonal antibodies (*Köhler and Milstein*)
- 1976 Overlapping genes in phage ϕ X174 (*Barrell, Air, Hutchinson*)
First transgenic mouse (*R. Jaenisch*)
Loci for structural genes on each human chromosome known (*Baltimore Conference on Human Gene Mapping*)
- 1977 Genes contain coding and noncoding DNA segments ("split genes") (exon/intron structure) (*R. J. Roberts, P. A. Sharp*, independently)
First recombinant DNA molecule that contains mammalian DNA
Methods to sequence DNA (*F. Sanger; Maxam and Gilbert*)
X-ray diffraction analysis of nucleosomes (*Finch et al.*)
- 1978 β -Globulin gene structure (*Leder, Weissmann, Tilghman and others*)
Mechanisms of transposition in bacteria
Production of somatostatin with recombinant DNA
Introduction of "chromosome walking" to find genes

- 1979 First genetic diagnosis using DNA technology (*Y. H. Kan*)
- 1980 Genes for embryonic development in *Drosophila* studied by mutational screen (*C. Nüsslein-Volhard and others*)
- 1981 Sequencing of a mitochondrial genome (*S. Anderson, S. G. Barrell, A. T. Bankier*)
- 1982 Tumor suppressor genes (*H. P. Klinger*)
Prions (proteinaceous infectious particles) proposed as cause of some chronic progressive central nervous system diseases (kuru, scrapie, Creutzfeldt-Jakob disease) (*S. B. Prusiner*)
- 1983 Cellular oncogenes (*H. E. Varmus and others*)
HIV virus (*L. Montagnier, R. Gallo*)
- 1984 Localization of the gene for Huntington disease (*Gusella*)
Identification of the T-cell receptor (*Tonegawa*)
Variable DNA sequences as "genetic fingerprints" (*A. Jeffreys*)
Helicobacter pylori (*B. Marshall*)
- 1985 Polymerase chain reaction (*Mullis, Saiki*)
Characterization of the gene for clotting factor VIII (*Gitschier*)
Sequencing of the AIDS virus
Localization of the gene for cystic fibrosis
Hypervariable DNA segments
Genomic imprinting in the mouse (*B. Cattanach*)
- 1986 First cloning of human genes. First identification of a human gene based on its chromosomal location (positional cloning) (*Royer-Pokora et al.*)
RNA as catalytic enzyme (*T. Cech*)
- 1987 Fine structure of an HLA molecule (*Björkman, Strominger et al.*)
Cloning of the gene for Duchenne muscular dystrophy (*Kunkel*)
Knockout mouse (*M. Capecchi*)
A genetic map of the human genome (*H. Donis-Keller et al.*)
Mitochondrial DNA and human evolution (*R. L. Cann, M. Stoneking, A. C. Wilson*)
- 1988 Start of the Human Genome Project
Successful gene therapy in vitro
Molecular structure of telomeres at the ends of chromosomes (*E. Blackburn and others*)
- 1989 Cloning of a defined region of a human chromosome obtained by microdissection (*Lüdecke, Senger, Claussen, Horsthemke*)
- 1990 Evidence for a defective gene causing inherited breast cancer (*Mary-Claire King*)
- 1991 Cloning of the gene for cystic fibrosis and for Duchenne muscular dystrophy
Odorant receptor multigene family (*Buck and Axel*)
Complete sequence of a yeast chromosome
Increasing use of microsatellites as polymorphic DNA markers
- 1992 Trinucleotide repeat expansion as a new class of human pathogenic mutations
High density map of DNA markers on human chromosomes
X chromosome inactivation center identified
p53 knockout mouse (*O. Smithies*)
- 1993 Gene for Huntington disease cloned
- 1994 Physical map of the human genome in high resolution
Mutations in fibroblast growth factor receptor genes as cause of achondroplasia and other human diseases
Identification of genes for breast cancer
- 1995 Master gene of the vertebrate eye, *sey* (small-eye) (*W. J. Gehring*)
STS-band map of the human genome (*T. J. Hudson et al.*)
- 1996 Yeast genome sequenced
Mouse genome map with more than 7000 markers (*E. S. Lander*)
- 1997 Sequence of *E. coli* (*F. R. Blattner et al.*), *Helicobacter pylori* (*J. F. Tomb*)
Mammal cloned by transfer of an adult cell nucleus into an enucleated oocyte (*Wilmut*)
Embryonic stem cells

- 1998 Nematode *C. elegans* genome sequenced (For more complete information, see references and <http://www.britannica.com> and click on the science channel, Lander and Weinberg, 2000.)
- 1999 First human chromosome (22) sequenced
Ribosome crystal structure
- 2000 *Drosophila* genome sequenced
First draft of the complete sequence of the human genome
First complete plant pathogen (*Xylella fastidiosa*) genome sequence
Arabidopsis thaliana, the first plant genome sequenced

Fundamentals

Molecular Basis of Genetics

The Cell and Its Components

Cells are the smallest organized structural units able to maintain an individual, albeit limited, life span while carrying out a wide variety of functions. Cells have evolved on earth during the past 3.5 billion years, presumably originating from suitable early molecular aggregations. Each cell originates from another living cell as postulated by R. Virchow in 1855 ("*omnis cellula e cellula*"). The living world consists of two basic types of cells: *prokaryotic cells*, which carry their functional information in a circular genome without a nucleus, and *eukaryotic cells*, which contain their genome in individual chromosomes in a nucleus and have a well-organized internal structure. Cells communicate with each other by means of a broad repertoire of molecular signals. Great progress has been made since 1839, when cells were first recognized as the "elementary particles of organisms" by M. Schleiden and T. Schwann. Today we understand most of the biological processes of cells at the molecular level.

A. Eukaryotic cells

A eukaryotic cell consists of cytoplasm and a nucleus. It is enclosed by a plasma membrane. The cytoplasm contains a complex system of inner membranes that form cellular structures (organelles). The main organelles are the mitochondria (in which important energy-delivering chemical reactions take place), the endoplasmic reticulum (consisting of a series of membranes in which glycoproteins and lipids are formed), the Golgi apparatus (for certain transport functions), and peroxisomes (for the formation or degradation of certain substances). Eukaryotic cells contain lysosomes, in which numerous proteins, nucleic acids, and lipids are broken down. Centrioles, small cylindrical particles made up of microtubules, play an essential role in cell division. Ribosomes are the sites of protein synthesis.

B. Nucleus of the Cell

The eukaryotic cell nucleus contains the genetic information. It is enclosed by an inner and an outer membrane, which contain pores for the transport of substances between the nucleus and the cytoplasm. The nucleus contains a

nucleolus and a fibrous matrix with different DNA-protein complexes.

C. Plasma membrane of the cell

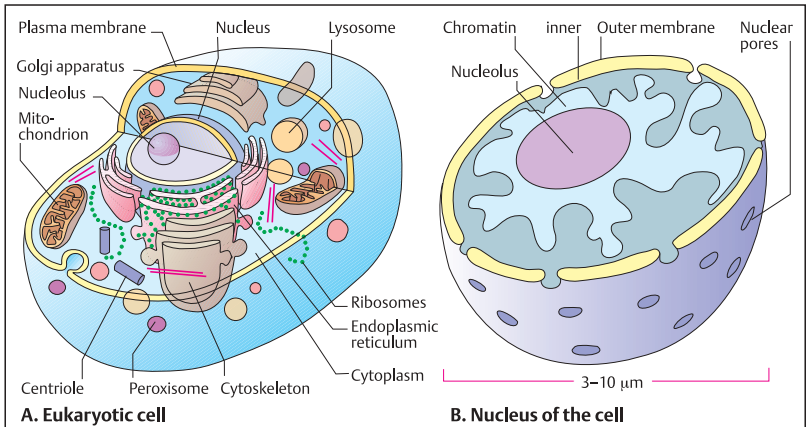
The environment of cells, whether blood or other body fluids, is water-based, and the chemical processes inside a cell involve water-soluble molecules. In order to maintain their integrity, cells must prevent water and other molecules from flowing in or out uncontrolled. This is accomplished by a water-resistant membrane composed of bipartite molecules of fatty acids, the *plasma membrane*. These molecules are phospholipids arranged in a double layer (bilayer) with a fatty interior. The plasma membrane itself contains numerous molecules that traverse the lipid bilayer once or many times to perform special functions. *Different types of membrane proteins* can be distinguished: (i) transmembrane proteins used as channels for transport of molecules into or out of the cell, (ii) proteins connected with each other to provide stability, (iii) receptor molecules involved in signal transduction, and (iv) molecules with enzyme function to catalyze internal chemical reactions in response to an external signal. (Figure redrawn from Alberts et al., 1998.)

D. Comparison of animal and plant cells

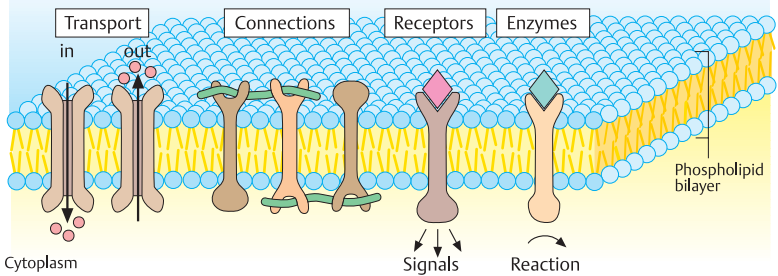
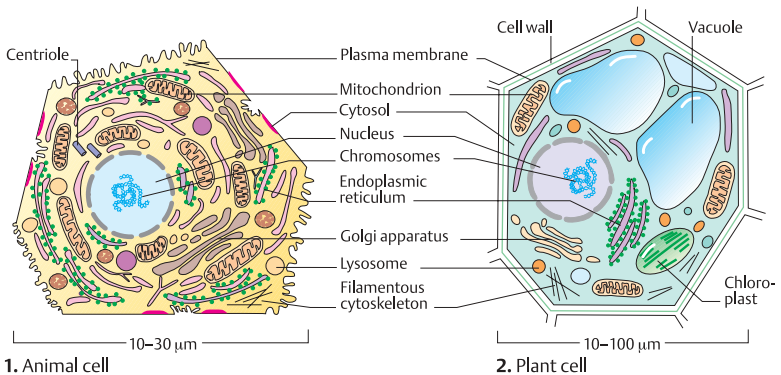
Plant and animal cells have many similar characteristics. One fundamental difference is that plant cells contain chloroplasts for photosynthesis. In addition, plant cells are surrounded by a rigid wall of cellulose and other polymeric molecules and contain vacuoles for water, ions, sugar, nitrogen-containing compounds, or waste products. Vacuoles are permeable to water but not to the other substances enclosed in the vacuoles. (Figures in A, B and D adapted from de Duve, 1984.)

References

- Alberts, B. et al.: *Essential Cell Biology. An Introduction to the Molecular Biology of the Cell*. Garland Publishing Co., New York, 1998.
- de Duve, C.: *A Guided Tour of the Living Cell*. Vol. I and II. Scientific American Books, Inc., New York, 1984.
- Lodish, H. et al.: *Molecular Cell Biology (with an animated CD-ROM)*. 4th ed. W.H. Freeman & Co., New York, 2000.



Extracellular space

**C. Plasma membrane****D. Comparison of the cell in animals and plants**

Some Types of Chemical Bonds

Close to 99% of the weight of a living cell is composed of just four elements: carbon (C), hydrogen (H), nitrogen (N), and oxygen (O). Almost 50% of the atoms are hydrogen atoms; about 25% are carbon, and 25% oxygen. Apart from water (about 70% of the weight of the cell) almost all components are carbon compounds. Carbon, a small atom with four electrons in its outer shell, can form four strong covalent bonds with other atoms. But most importantly, carbon atoms can combine with each other to build chains and rings, and thus large complex molecules with specific biological properties.

A. Compounds of hydrogen (H), oxygen (O), and carbon (C)

Four simple combinations of these atoms occur frequently in biologically important molecules: hydroxyl ($-\text{OH}$; alcohols), methyl ($-\text{CH}_3$), carboxyl ($-\text{COOH}$), and carbonyl ($\text{C}=\text{O}$; aldehydes and ketones) groups. They impart to the molecules characteristic chemical properties, including possibilities to form compounds.

B. Acids and esters

Many biological substances contain a carbon-oxygen bond with weak acidic or basic (alkaline) properties. The degree of acidity is expressed by the pH value, which indicates the concentration of H^+ ions in a solution, ranging from 10^{-1} mol/L (pH 1, strongly acidic) to 10^{-14} mol/L (pH 14, strongly alkaline). Pure water contains 10^{-7} moles H^+ per liter (pH 7.0). An ester is formed when an acid reacts with an alcohol. Esters are frequently found in lipids and phosphate compounds.

C. Carbon–nitrogen bonds (C–N)

C–N bonds occur in many biologically important molecules: in amino groups, amines, and amides, especially in proteins. Of paramount significance are the amino acids (cf. p. 30), which are the subunits of proteins. All proteins have a specific role in the functioning of an organism.

D. Phosphate compounds

Ionized phosphate compounds play an essential biological role. HPO_4^{2-} is a stable inorganic phosphate ion from ionized phosphoric acid. A phosphate ion and a free hydroxyl group can

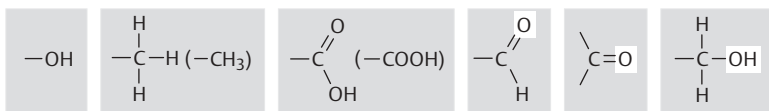
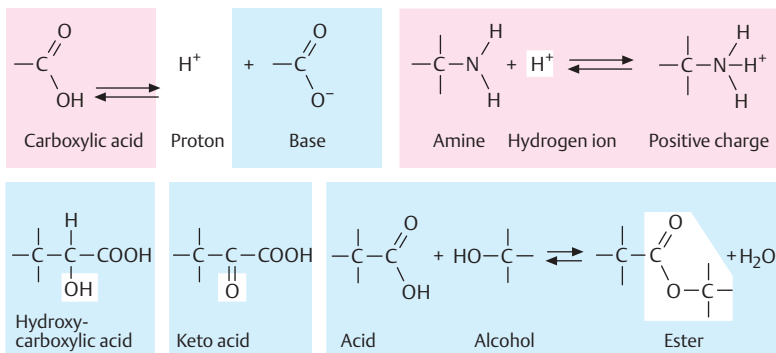
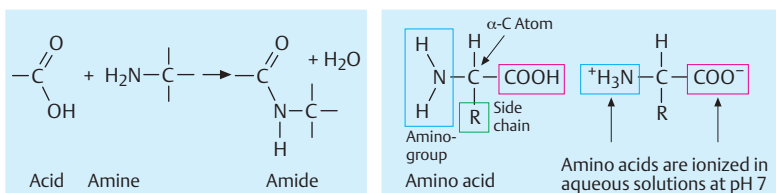
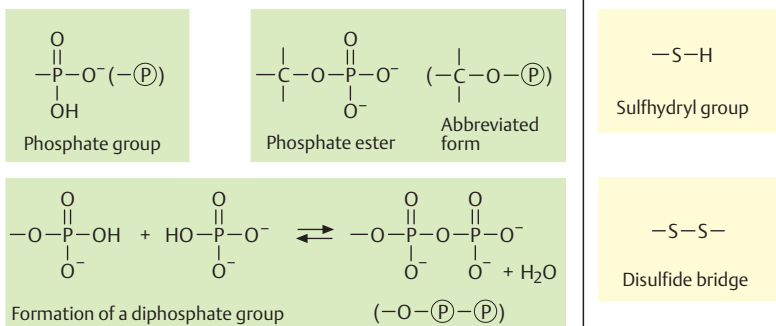
form a phosphate ester. Phosphate compounds play an important role in energy-rich molecules and numerous macromolecules because they can store energy.

E. Sulfur compounds

Sulfur often serves to bind biological molecules together, especially when two sulfhydryl groups ($-\text{SH}$) react to form a disulfide bridge ($-\text{S}-\text{S}-$). Sulfur is a component of two amino acids (cysteine and methionine) and of some polysaccharides and sugars. Disulfide bridges play an important role in many complex molecules, serving to stabilize and maintain particular three-dimensional structures.

References

- Alberts, B. et al.: *Molecular Biology of the Cell*. 3rd ed. Garland Publishing Co., New York, 1994.
- Koolman, J., Röhm K.H.: *Color Atlas of Biochemistry*. Thieme, Stuttgart – New York, 1996.
- Stryer, L.: *Biochemistry*, 4th ed. W.H. Freeman & Co., New York, 1995.

**A. Functional groups with hydrogen (H), oxygen (O), and carbon (C)****B. Acids and esters****C. Carbon-nitrogen compounds (C—N)****D. Phosphate compounds****E. Sulfur groups**

Carbohydrates

Carbohydrates in their various chemical forms and their derivatives are an important group of biomolecules for genetics. They provide the basic framework of DNA and RNA. Their flexibility makes them especially suitable for transferring genetic information from cell to cell.

Along with nucleic acids, lipids, and proteins, carbohydrates are one of the most important classes of biomolecules. Their main functions can be classified into three groups: (i) to deliver and store energy, (ii) to help form DNA and RNA, the information-carrying molecules (see pp. 34 and 38), (iii) to help form cell walls of bacteria and plants. Carbohydrates are often bound to proteins and lipids.

As polysaccharides, carbohydrates are important structural elements of the cell walls of animals, bacteria, and plants. They form cell surface structures (receptors) used in conducting signals from cell to cell. Combined with numerous proteins and lipids, carbohydrates are important components of numerous cell structures. Finally, they function to transfer and store energy in intermediary metabolism.

A. Monosaccharides

Monosaccharides (simple sugars) are aldehydes ($-C=O$, $-H$) or ketones ($>C=O$) with two or more hydroxy groups (general structural formula $(CH_2O)_n$). The aldehyde or ketone group can react with one of the hydroxy groups to form a ring. This is the usual configuration of sugars that have five or six C atoms (pentoses and hexoses). The C atoms are numbered. The D- and the L-forms of sugars are mirror-image isomers of the same molecule.

The naturally occurring forms are the D-(dextro) forms. These further include β - and α -forms as stereoisomers. In the cyclic forms the C atoms of sugars are not on a plane, but three-dimensionally take the shape of a chair or a boat. The β -D-glucopyranose configuration (glucose) is the energetically favored, since all the axial positions are occupied by H atoms. The arrangement of the $-OH$ groups can differ, so that stereoisomers such as mannose or galactose are formed.

B. Disaccharides

These are compounds of two monosaccharides. The aldehyde or ketone group of one can bind to

an α -hydroxy or a β -hydroxy group of the other. Sucrose and lactose are frequently occurring disaccharides.

C. Derivatives of sugars

When certain hydroxy groups are replaced by other groups, sugar derivatives are formed. These occur especially in polysaccharides. In a large group of genetically determined syndromes, complex polysaccharides can not be degraded owing to reduced or absent enzyme function (mucopolysaccharidoses, mucopolysaccharidoses) (see p. 356).

D. Polysaccharides

Short (oligosaccharides) and long chains of sugars and sugar derivatives (polysaccharides) form essential structural elements of the cell. Complex oligosaccharides with bonds to proteins or lipids are part of cell surface structures, e.g., blood group antigens.

Examples of human hereditary disorders in the metabolism of carbohydrates

Diabetes mellitus: a heterogeneous group of disorders characterized by elevated levels of blood glucose, with complex clinical and genetic features (see p. 362).

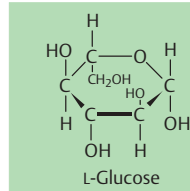
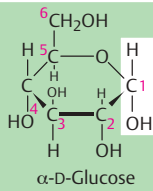
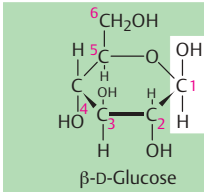
Disorders of fructose metabolism: Three inherited disorders are known: benign fructosemia, hereditary fructose intolerance with hypoglycemia and vomiting, and hereditary fructose 1,6-bisphosphate deficiency with hypoglycemia, apnea, lactic acidosis, and often lethal outcome in newborn infants.

Glycogen storage diseases: a group of disorders of glycogen metabolism that differ in clinical symptoms and the genes and enzymes involved.

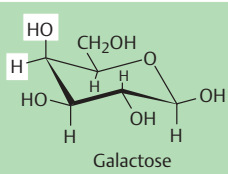
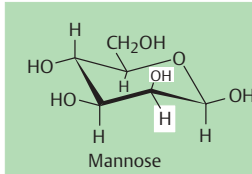
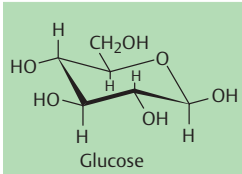
Galactose metabolism: Three different inherited disorders with acute toxicity and long-term effects.

References

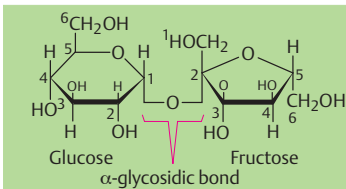
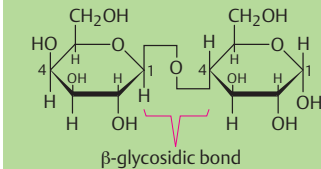
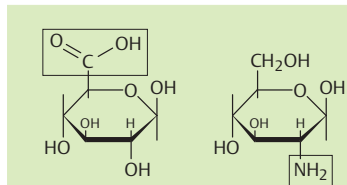
- Gilbert-Barness, E., Barness, L.: Metabolic Diseases. Foundations of Clinical Management, Genetics, and Pathology. Eaton Publishing, Natick, MA 01760, USA, 2000.
- Scriver, C. R., Beaudet, A. L., Sly, W. S., Valle, D., editors: The Metabolic and Molecular Bases of Inherited Disease. 8th ed., McGraw-Hill, New York, 2001.



Stereoisomers

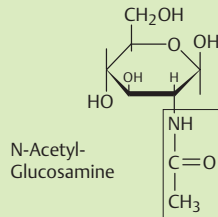
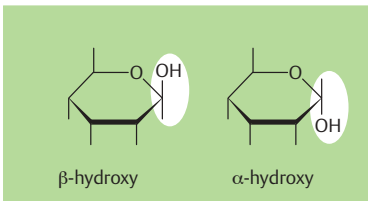
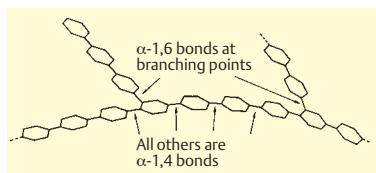


Isomers of glucose

A. MonosaccharidesSucrose (Glucose- α -1, 2-fructose)Lactose (Galactose- β -1.4-glucose)

Glucuronic acid

Glucosamine

**C. Sugar derivatives****B. Disaccharides****D. Polysaccharides**

Lipids (Fats)

Lipids usually occur as large molecules (macromolecules). They are essential components of membranes and precursors of other important biomolecules, such as steroids for the formation of hormones and other molecules for transmitting intercellular signals. In addition to fatty acids, compounds with carbohydrates (glycolipids), phosphate groups (phospholipids), and other molecules are especially important. A special characteristic is their pronounced polarity, with a hydrophilic (water-attracting) and a hydrophobic (water-repelling) region. This makes lipids especially suited for forming the outer limits of the cell (cell membrane).

A. Fatty acids

Fatty acids are composed of a hydrocarbon chain with a terminal carboxylic acid group. Thus, they are polar, with a hydrophilic ($-\text{COOH}$) and a hydrophobic end ($-\text{CH}_3$), and differ in the length of the chain and its degree of saturation. When one or more double bonds occur in the chain, the fatty acid is referred to as unsaturated. A double bond makes the chain relatively rigid and causes a kink. Fatty acids form the basic framework of many important macromolecules. The free carboxyl group ($-\text{COOH}$) of a fatty acid is ionized ($-\text{COO}^-$).

B. Lipids

Fatty acids can combine with other groups of molecules to form other types of lipids. As water-insoluble (hydrophobic) molecules, they are soluble only in organic solvents. The carboxyl group can enter into an ester or an amide bond. Triglycerides are compounds of fatty acids with glycerol.

Glycolipids (lipids with sugar residues) and phospholipids (lipids with a phosphate group attached to an alcohol derivative) are the structural bases of important macromolecules. Their intracellular degradation requires the presence of numerous enzymes, disorders of which have a genetic basis and lead to numerous genetically determined diseases.

Sphingolipids are an important group of molecules in biological membranes. Here, sphingosine, instead of glycerol, is the fatty acid-binding molecule. Sphingomyelin and gangliosides contain sphingosine. Gangliosides make up 6% of the central nervous system

lipids. They are degraded by a series of enzymes. Genetically determined disorders of their catabolism lead to severe diseases, e.g., Tay-Sachs disease due to defective degradation of ganglioside GM2 (deficiency of β -N-acetylhexosaminidase).

C. Lipid aggregates

Owing to their bipolar properties, fatty acids can form lipid aggregates in water. The hydrophilic ends are attracted to their aqueous surroundings; the hydrophobic ends protrude from the surface of the water and form a surface film. If completely under the surface, they may form a micelle, compact and dry within. Phospholipids and glycolipids can form two-layered membranes (lipid membrane bilayer). These are the basic structural elements of cell membranes, which prevent molecules in the surrounding aqueous solution from invading the cell.

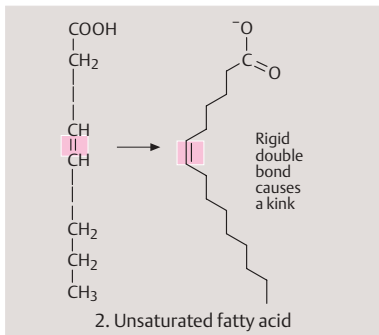
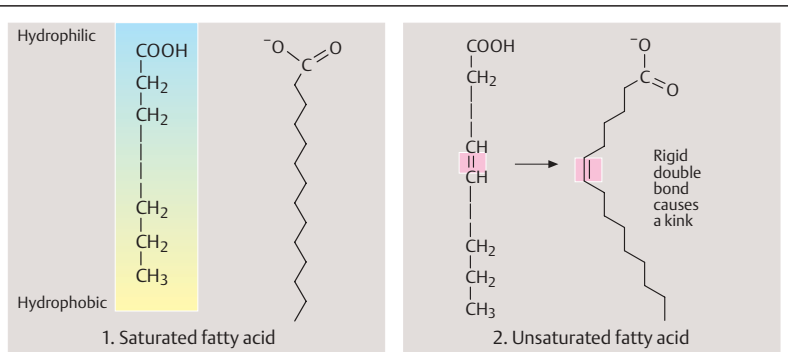
D. Other lipids: steroids

Steroids are small molecules consisting of four different rings of carbon atoms. Cholesterol is the precursor of five major classes of steroid hormones: prostagens, glucocorticoids, mineralocorticoids, androgens, and estrogens. Each of these hormone classes is responsible for important biological functions such as maintenance of pregnancy, fat and protein metabolism, maintenance of blood volume and blood pressure, and development of sex characteristics.

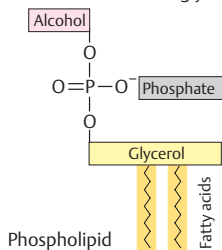
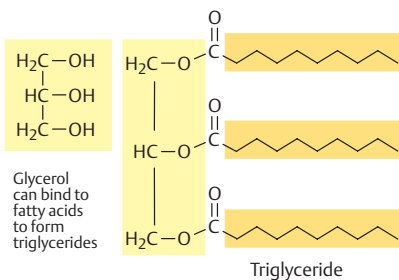
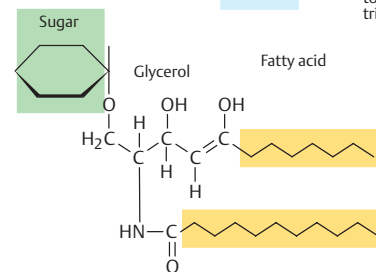
Examples of human hereditary disorders in lipoprotein and lipid metabolism

Scriver et al. (2001) list several groups of disorders. Important examples are familial hypercholesterolemia (see 358), familial lipoprotein lipase deficiency, dysbetalipoproteinemia, and disorders of high-density lipoprotein.

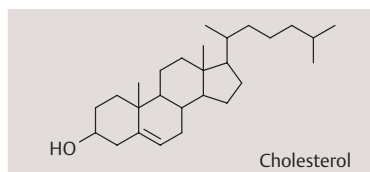
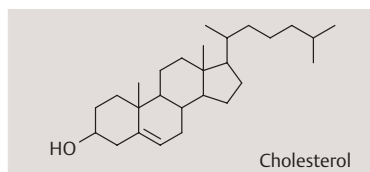
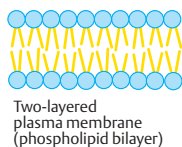
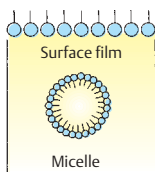
(Scriver et al., 2001; Gilbert-Barness & Barness, 2000).



A. Fatty acids



B. Lipids



C. Lipid aggregates

D. Other lipids: steroids

Nucleotides and Nucleic Acids

Nucleotides participate in almost all biological processes. They are the subunits of DNA and RNA, the molecules that carry genetic information (see p. 34). Nucleotide derivatives are involved in the biosynthesis of numerous molecules; they convey energy, are part of essential coenzymes, and regulate numerous metabolic functions. Since all these functions are based on genetic information of the cells, nucleotides represent a central class of molecules for genetics. Nucleotides are composed of three integral parts: phosphates, sugars, and purine or pyrimidine bases.

A. Phosphate groups

Phosphate groups may occur alone (monophosphates), in twos (diphosphates) or in threes (triphosphates). They are normally bound to the hydroxy group of the C atom in position 5 of a five-C-atom sugar (pentose).

B. Sugar residues

The sugar residues in nucleotides are usually derived from either ribose (in ribonucleic acid, RNA) or deoxyribose (in deoxyribonucleic acid, DNA) (ribonucleoside or deoxyribonucleoside). These are the base plus the respective sugar.

C. Nucleotide bases of pyrimidine

Cytosine (C), thymine (T), and uracil (U) are the three pyrimidine nucleotide bases. They differ from each other in their side chains ($-\text{NH}_2$ on C4 in cytosine, $-\text{CH}_3$ on C5 in thymine, O on C4 in uracil) and in the presence or absence of a double bond between N3 and C4 (present in cytosine).

D. Nucleotide bases of purine

Adenine (A) and guanine (G) are the two nucleotide bases of purine. They differ in their side chains and a double bond (between N1 and C6).

E. Nucleosides

A nucleoside is a compound of a sugar residue (ribose or deoxyribose) and a nucleotide base. The bond is between the C atom in position 1 of the sugar (as in compounds of sugars) and an N atom of the base (N-glycosidic bond). The nucleotides of the various bases are named according to whether they are a ribonucleoside or a deoxyribonucleoside, e.g., adenosine or

deoxyadenosine, guanosine or deoxyguanosine, uridine (occurs only as a ribonucleoside), cytidine or deoxycytidine. Thymidine occurs only as a deoxynucleoside.

F. Nucleotides

A nucleotide is a compound of a five-C-atom sugar residue (ribose or deoxyribose) attached to a nucleotide base (pyrimidine or purine base) and a phosphate group. Nucleotides are the subunits of nucleic acids. The nucleotides of the individual bases are referred to as follows: adenylate (AMP, adenosine monophosphate), guanylate monophosphate (GMP), uridylylate (UMP), and cytidylate (CMP) for the ribonucleotides (5' monophosphates) and deoxyadenylate (dAMP), deoxyguanylate (dGMP), deoxythymidylate (dTMP), and deoxycytidylate (dCMP) for the deoxyribonucleotides.

G. Nucleic acids

Nucleic acids are formed when nucleotides are joined to each other by means of phosphodiester bridges between the 3' C atom of one nucleotide and the 5' C atom of the next. The linear sequence of nucleotides is usually given in the 5' to 3' direction with the abbreviations of the respective nucleotide bases. For instance, ATCG would signify the sequence adenine (A), thymine (T), cytosine (C), and guanine (G) in the 5' to 3' direction.

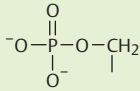
Examples of human hereditary disorders in purine and pyrimidine metabolism

Hyperuricemia and gout: A group of disorders resulting from genetically determined excessive synthesis of purine precursors.

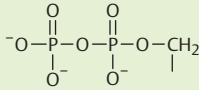
Lesch-Nyhan syndrome: A variable, usually severe infantile X-chromosomal disease with marked neurological manifestations resulting from hypoxanthine-guanine phosphoribosyl-transferase deficiency.

Adenosine deaminase deficiency: A heterogeneous group of disorders resulting in severe infantile immunodeficiency. Different autosomal recessive and X-chromosomal types exist.

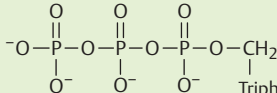
Scriver et al., 2001; Gilbert-Barness & Barness, 2000).



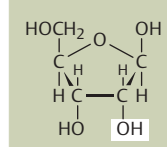
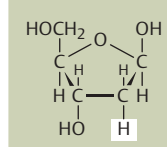
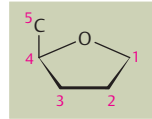
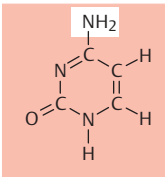
Monophosphate



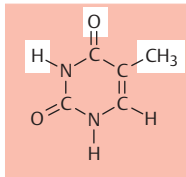
Diphosphate



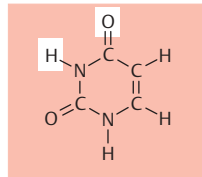
Triphosphate

 β -D-Ribose β -D-Deoxyribose**A. Phosphate groups****B. Sugar residues (pentose)**

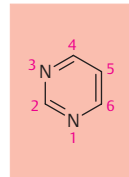
Cytosine (C)



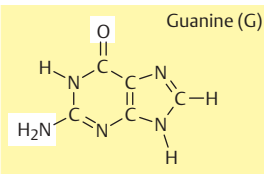
Thymine (T)



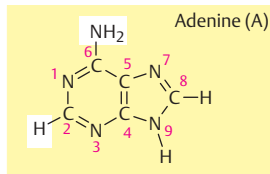
Uracil (U)



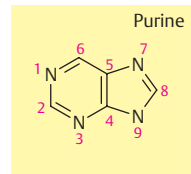
Pyrimidine

C. Nucleotide bases of pyrimidine

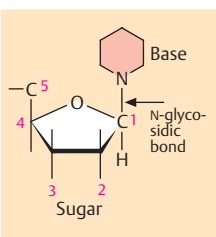
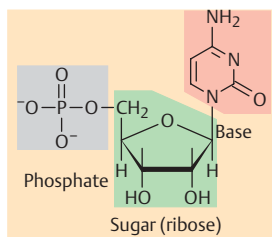
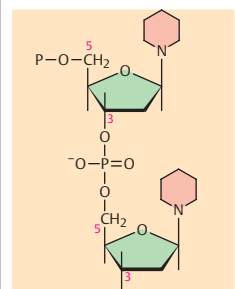
Guanine (G)



Adenine (A)



Purine

D. Nucleotide bases of purine**E. Nucleoside**
(base and sugar)**F. Nucleotide**
(base + sugar + phosphate)**G. Nucleic acid**

Amino Acids

Amino acids are the basic structural units of proteins. A defined linear sequence of the amino acids and a specific three-dimensional structure confer quite specific physicochemical properties to each protein. An amino acid consists of a “central” carbon with one bond to an amino group ($-\text{NH}_2$) one to a carboxyl group ($-\text{COOH}$) one to a hydrogen atom, and the fourth to a variable side chain. Amino acids are ionized in neutral solutions, since the amino group takes on a proton ($-\text{NH}_3^+$) and the carboxyl group dissociates ($-\text{COO}^-$). The side chain determines the distinguishing characteristics of an amino acid, including the size, form, electrical charge or hydrogen-bonding ability, and the total specific chemical reactivity. Amino acids can be differentiated according to whether they are neutral or not neutral (basic or acidic) and whether they have a polar or nonpolar side chain. Each amino acid has its own three-letter and one-letter abbreviations. Essential amino acids in vertebrates are His, Ile, Leu, Lys, Met, Phe, Thr, Tyr, and Val.

A. Neutral amino acids, nonpolar side chains

All neutral amino acids have a $-\text{COO}^-$ and an $-\text{NH}_3^+$ group. The simplest amino acids have a simple aliphatic side chain. For glycine this is merely a hydrogen atom ($-\text{H}$); for alanine it is a methyl group ($-\text{CH}_3$). Larger side chains occur on valine, leucine, and isoleucine. These larger side chains are hydrophobic (water-repellent) and make their respective amino acids less water-soluble than do hydrophilic (water-attracting) chains. Proline has an aliphatic side chain that, unlike in other amino acids, is bound to both the central carbon and to the amino group, so that a ringlike structure is formed. Aromatic side chains occur in phenylalanine (a phenyl group bound via a methylene group ($-\text{CH}_2-$) and tryptophan (an indol ring bound via a methylene group). These amino acids are very hydrophobic. Two amino acids contain sulfur (S) atoms. In cysteine this is in the form of a sulfhydryl group ($-\text{SH}$); in methionine it is a thioether ($-\text{S}-\text{CH}_3$). Both are hydrophobic. The sulfhydryl group in cysteine is very reactive and participates in forming disulfide bonds ($-\text{S}-\text{S}-$). These play an important role in stabilizing the three-dimensional forms of proteins.

B. Hydrophilic amino acids, polar side chains

Serine, threonine, and tyrosine contain hydroxyl groups ($-\text{OH}$). Thus, they are hydrolyzed forms of glycine, alanine, and phenylalanine. The hydroxyl groups make them hydrophilic and more reactive than the nonhydrolyzed forms. Asparagine and glutamine both contain an amino and an amide group. At physiological pH their side chains are negatively charged.

C. Charged amino acids

These amino acids have either two ionized amino groups (basic) or two carboxyl groups (acidic). Basic amino acids (positively charged) are arginine, lysine, and histidine. Histidine has an imidazole ring and can be uncharged or positively charged, depending on its surroundings. It is frequently found in the reactive centers of proteins, where it takes part in alternating bonds (e.g., in the oxygen-binding region of hemoglobin). Aspartic acid and glutamic acid each have two carboxyl groups ($-\text{COOH}$) and are thus (as a rule) acidic.

Seven of the 20 amino acids have slightly ionizable side chains, making them highly reactive (Asn, Glu, His, Cys, Tyr, Lys, Arg).

Examples of human hereditary disorders of amino acid metabolism

The amino acids glycine, phenylalanine, tyrosine, histidine, proline, lysine, and the branched chain amino acids valine, leucine, and isoleucine are predominantly involved in various disorders leading to toxic metabolic symptoms due to an increase or decrease of their plasma concentration.

Phenylketonuria: Disorders of phenylalanine hydroxylation result in variable clinical signs depending on the severity, caused by a spectrum of mutations in the responsible gene.

Maple syrup urine disease: A variable disorder due to deficiency of branched chain α -keto acid dehydrogenase, which leads to accumulation of valine, leucine, and isoleucine. The classic severe form results in severe neurological damage to the infant.

Scriver et al., 2001; Gilbert-Barness & Barness, 2000).

$\begin{array}{c} \text{COO}^- \\ \\ \text{H}_3\text{N}^+-\text{C}-\text{H} \\ \\ \text{H} \end{array}$ Glycine Gly (G)	$\begin{array}{c} \text{COO}^- \\ \\ \text{H}_3\text{N}^+-\text{C}-\text{H} \\ \\ \text{CH}_3 \end{array}$ Alanine Ala (A)	$\begin{array}{c} \text{COO}^- \\ \\ \text{H}_3\text{N}^+-\text{C}-\text{H} \\ \\ \text{CH}-\text{CH}_3 \\ \\ \text{CH}_3 \end{array}$ Valine Val (V)	$\begin{array}{c} \text{COO}^- \\ \\ \text{H}_3\text{N}^+-\text{C}-\text{H} \\ \\ \text{CH}_2 \\ \\ \text{CH}-\text{CH}_3 \\ \\ \text{CH}_3 \end{array}$ Leucine Leu (L)	$\begin{array}{c} \text{COO}^- \\ \\ \text{H}_3\text{N}^+-\text{C}-\text{H} \\ \\ \text{CH}-\text{CH}_2 \\ \quad \\ \text{CH}_3 \quad \text{CH}_3 \end{array}$ Isoleucine Ile (I)
$\begin{array}{c} \text{COO}^- \\ \\ \text{H}_2\text{N}^+-\text{C}_4\text{H}_7 \end{array}$ Proline Pro (P)	$\begin{array}{c} \text{COO}^- \\ \\ \text{H}_3\text{N}^+-\text{C}-\text{H} \\ \\ \text{CH}_2 \\ \\ \text{C}_6\text{H}_5 \end{array}$ Phenylalanine Phe (F)	$\begin{array}{c} \text{COO}^- \\ \\ \text{H}_3\text{N}^+-\text{C}-\text{H} \\ \\ \text{CH} \\ \\ \text{CH}_2 \\ \\ \text{C}=\text{CH} \\ \quad \\ \text{C}_6\text{H}_4 \quad \text{NH} \end{array}$ Tryptophan Trp (W)	$\begin{array}{c} \text{COO}^- \\ \\ \text{H}_3\text{N}^+-\text{C}-\text{H} \\ \\ \text{CH}_2 \\ \\ \text{SH} \end{array}$ Cysteine Cys (C)	$\begin{array}{c} \text{COO}^- \\ \\ \text{H}_3\text{N}^+-\text{C}-\text{H} \\ \\ \text{CH}_2 \\ \\ \text{CH}_2 \\ \\ \text{S} \\ \\ \text{CH}_3 \end{array}$ Methionine Met (M)

A. Neutral amino acids, nonpolar side chains

$\begin{array}{c} \text{COO}^- \\ \\ \text{H}_3\text{N}^+-\text{C}-\text{H} \\ \\ \text{CH}_2\text{OH} \end{array}$ Serine Ser (S)	$\begin{array}{c} \text{COO}^- \\ \\ \text{H}_3\text{N}^+-\text{C}-\text{H} \\ \\ \text{CHOH} \\ \\ \text{CH}_3 \end{array}$ Threonine Thr (T)	$\begin{array}{c} \text{COO}^- \\ \\ \text{H}_3\text{N}^+-\text{C}-\text{H} \\ \\ \text{CH}_2 \\ \\ \text{C}_6\text{H}_4 \\ \\ \text{OH} \end{array}$ Tyrosine Tyr (Y)	$\begin{array}{c} \text{COO}^- \\ \\ \text{H}_3\text{N}^+-\text{C}-\text{H} \\ \\ \text{CH}_2 \\ \\ \text{H}_2\text{N}-\text{C}=\text{O} \end{array}$ Asparagine Asn (N)	$\begin{array}{c} \text{COO}^- \\ \\ \text{H}_3\text{N}^+-\text{C}-\text{H} \\ \\ \text{CH}_2 \\ \\ \text{CH}_2 \\ \\ \text{H}_2\text{N}-\text{C}=\text{O} \end{array}$ Glutamine Gln (Q)
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B. Neutral amino acids, polar side chains

$\begin{array}{c} \text{COO}^- \\ \\ \text{H}_3\text{N}^+-\text{C}-\text{H} \\ \\ \text{CH}_2 \\ \\ \text{CH}_2 \\ \\ \text{CH}_2 \\ \\ \text{NH} \\ \\ \text{C}=\text{NH}_2^+ \\ \\ \text{H}_2\text{N} \end{array}$ Arginine Arg (R)	1. Basic (positively charged) $\begin{array}{c} \text{COO}^- \\ \\ \text{H}_3\text{N}^+-\text{C}-\text{H} \\ \\ \text{CH}_2 \\ \\ \text{CH}_2 \\ \\ \text{CH}_2 \\ \\ \text{CH}_2=\text{NH}_3^+ \end{array}$ Lysine Lys (K)	$\begin{array}{c} \text{COO}^- \\ \\ \text{H}_3\text{N}^+-\text{C}-\text{H} \\ \\ \text{CH}_2 \\ \\ \text{C}_3\text{H}_3\text{N} \end{array}$ Histidine His (H)	2. Acid (negatively charged) $\begin{array}{c} \text{COO}^- \\ \\ \text{H}_3\text{N}^+-\text{C}-\text{H} \\ \\ \text{CH}_2 \\ \\ \text{COO}^- \end{array}$ Aspartic acid Asp (D)	$\begin{array}{c} \text{COO}^- \\ \\ \text{H}_3\text{N}^+-\text{C}-\text{H} \\ \\ \text{CH}_2 \\ \\ \text{CH}_2 \\ \\ \text{COO}^- \end{array}$ Glutamic acid Glu (E)
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C. Charged amino acids

Proteins

Proteins are involved in practically all chemical processes in living organisms. Their universal significance is apparent in that, as enzymes, they drive chemical reactions in living cells. Without enzymatic catalysis, the macromolecules involved would not react spontaneously. All enzymes are the products of one or more genes. Proteins also serve to transport small molecules, ions, or metals. Proteins have important functions in cell division during growth and in cell and tissue differentiation. They control the coordination of movements by regulating muscle cells and the production and transmission of impulses within and between nerve cells. They control blood homeostasis (blood clotting) and immune defense. They carry out mechanical functions in skin, bone, blood vessels, and other areas.

A. Joining of amino acids (peptide bonds)

The basic units of proteins, amino acids, can be joined together very easily owing to their dipolar ionization (zwitterions). The carboxyl group of one amino acid binds to the amino group of the next (a peptide bond, sometimes also referred to as an amide bond). When many amino acids are bound together by peptide bonds, they form a polypeptide chain. Each polypeptide chain has a defined direction, determined by the amino group ($-\text{NH}_2$) at one end and the carboxyl group ($-\text{COOH}$) at the other. By convention, the amino group represents the beginning, and the carboxyl group the end of a peptide chain.

B. Primary structure of a protein

The determination of the complete amino acid sequence of insulin by Frederick Sanger in 1955 was a landmark accomplishment. It showed for the first time that a protein, in genetic terms a gene product, has a precisely defined amino acid sequence. The amino acid sequence yields important information about the function and evolutionary origin of a protein.

The primary structure of a protein is its amino acid sequence in a one-dimensional plane. As are many other proteins, insulin is synthesized from precursor molecules: preproinsulin and proinsulin. Preproinsulin consists of 110 amino acids including 24 amino acids of a leader

sequence at the amino end. The leader sequence directs the molecule to the correct site in the cell and is then removed to yield proinsulin. This is converted to insulin by removal of the connecting peptide (C peptide) consisting of amino acids 31–65. Amino acids 1–30 form the B chain; the remaining (66–86) amino acids form the A chain. The A and the B chains are connected by two disulfide bridges joining the cysteines in position 7 and position 20 of the A chain to those of positions 7 and 19, respectively, of the B chain. The A chain contains a disulfide bridge between positions 6 and 11. The positions of the cysteines reflect the spatial arrangements of the amino acids, called the secondary structure.

C. Secondary structural units, the α helix and the β sheet

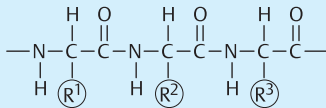
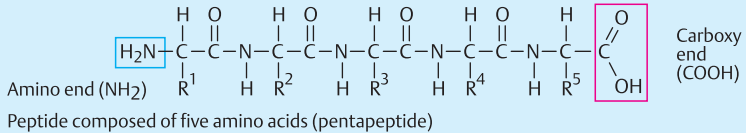
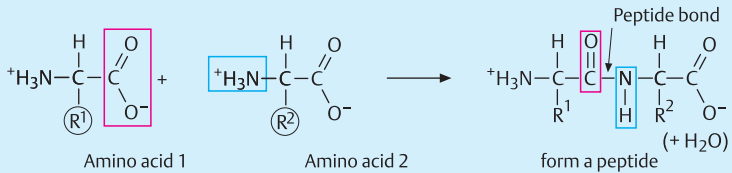
Two basic units of global proteins are α helix formation (α helix) and a flat sheet (β pleated sheet). Panel C shows a schematic drawing of a unit of one α helix between two β -sheets, called a $\beta\alpha\beta$ unit (Figure redrawn from Stryer, 1995).

D. Tertiary structure of insulin

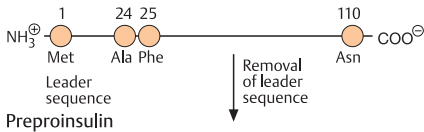
All functional proteins assume a well-defined three-dimensional structure. This structure is defined by the sequence of amino acids and their physicochemical properties. Tertiary structure is defined by the spatial arrangement of amino acid residues that are far apart in the linear sequence. The quaternary structure is the folding of the protein resulting in a specific three-dimensional spatial arrangement of the subunits and the nature of their contacts. The correct quaternary structure ensures proper function. (Figure from Koolman & Röhm, 1996).

References

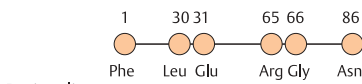
- Koolman, J., Röhm, K.-H.: Color Atlas of Biochemistry. Thieme, Stuttgart–New York, 1996.
- Stryer, L.: Biochemistry, 4th ed. W.H. Freeman & Co., New York, 1995.



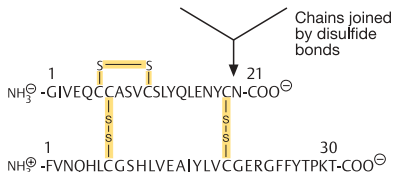
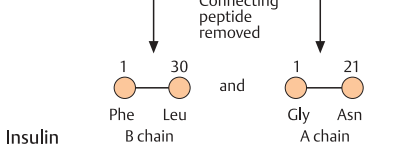
A. Joining of amino acids (peptide bond)



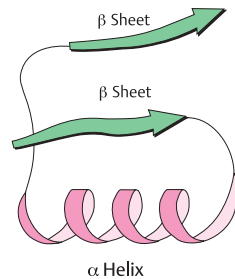
Preproinsulin



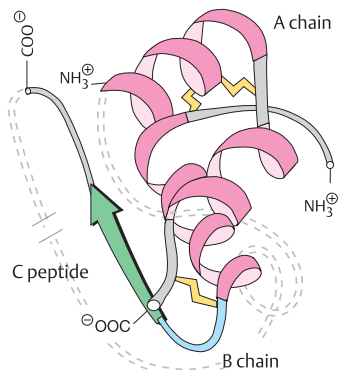
Proinsulin



B. Primary structure of a protein



C. Secondary structural units, the α helix and the β sheet



D. Tertiary structure of insulin

DNA as Carrier of Genetic Information

Although DNA was discovered in 1869 by Friedrich Miescher as a new, acidic, phosphorus-containing substance made up of very large molecules that he named "nuclein", its biological role was not recognized. In 1889 Richard Altmann introduced the term "nucleic acid". By 1900 the purine and pyrimidine bases were known. Twenty years later, the two kinds of nucleic acids, RNA and DNA, were distinguished. An incidental but precise observation (1928) and relevant investigations (1944) indicated that DNA could be the carrier of genetic information.

A. The observation of Griffith

In 1928 the English microbiologist Fred Griffith made a remarkable observation. While investigating various strains of *Pneumococcus*, he determined that mice injected with strain S (smooth) died (1). On the other hand, animals injected with strain R (rough) lived (2). When he inactivated the lethal S strain by heat, there were no sequelae, and the animal survived (3). Surprisingly, a mixture of the nonlethal R strain and the heat-inactivated S strain had a lethal effect like the S strain (4). And he found normal living pneumococci of the S strain in the animal's blood. Apparently, cells of the R strain were changed into cells of the S strain (transformed). For a time, this surprising result could not be explained and was met with skepticism. Its relevance for genetics was not apparent.

B. The transforming principle is DNA

Griffith's findings formed the basis for investigations by Avery, MacLeod, and McCarty (1944). Avery and co-workers at the Rockefeller Institute in New York elucidated the chemical basis of the transforming principle. From cultures of an S strain (1) they produced an extract of lysed cells (cell-free extract) (2). After all its proteins, lipids, and polysaccharides had been removed, the extract still retained the ability to transform pneumococci of the R strain to pneumococci of the S strain (transforming principle) (3). With further studies, Avery and co-workers determined that this was attributed to the DNA alone. Thus, the DNA must contain the corre-

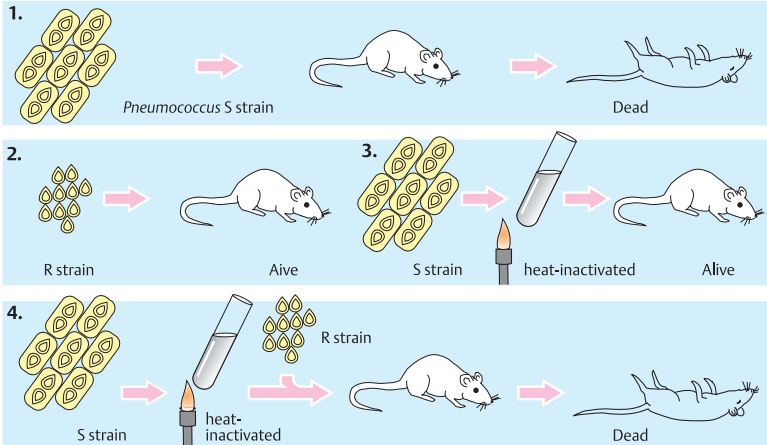
sponding genetic information. This explained Griffith's observation. Heat inactivation had left the DNA of the bacterial chromosomes intact. The section of the chromosome with the gene responsible for capsule formation (S gene) could be released from the destroyed S cells and be taken up by some R cells in subsequent cultures. After the S gene was incorporated into its DNA, an R cell was transformed into an S cell (4). Page 90 shows how bacteria can take up foreign DNA so that some of their genetic attributes will be altered correspondingly.

C. Genetic information is transmitted by DNA alone

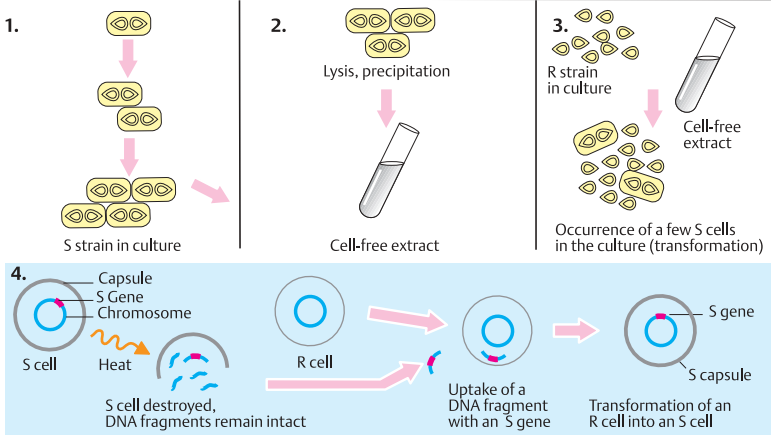
The final evidence that DNA, and no other molecule, transmits genetic information was provided by Hershey and Chase in 1952. They labeled the capsular protein of bacteriophages (see p. 88) with radioactive sulfur (^{35}S) and the DNA with radioactive phosphorus (^{32}P). When bacteria were infected with the labeled bacteriophage, only ^{32}P (DNA) entered the cells, and not the ^{35}S (capsular protein). The subsequent formation of new, complete phage particles in the cell proved that DNA was the exclusive carrier of the genetic information needed to form new phage particles, including their capsular protein. Next, the structure and function of DNA needed to be clarified. The genes of all cells and some viruses consist of DNA, a long-chained threadlike molecule.

References

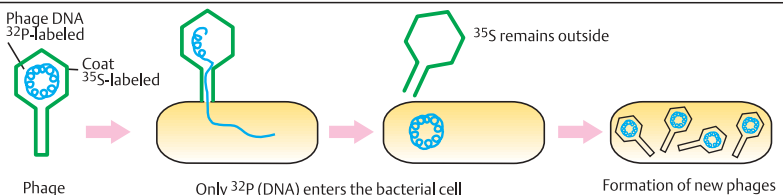
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A. The observation of Griffith



B. The transforming principle is DNA



C. Genetic information is transferred by DNA only

DNA and Its Components

The information for the development and specific functions of cells and tissues is stored in the genes. A gene is a portion of the genetic information, definable according to structure and function. Genes lie on chromosomes in the nuclei of cells. They consist of a complex long-chained molecule, deoxyribonucleic acid (DNA). In the following, the constituents of the DNA molecule will be presented. DNA is a nucleic acid. Its chemical components are nucleotide bases, a sugar (deoxyribose), and phosphate groups. They determine the three-dimensional structure of DNA, from which it derives its functional consequence.

A. Nucleotide bases

The nucleotide bases in DNA are heterocyclic molecules derived from either pyrimidine or purine. Five bases occur in the two types of nucleic acids, DNA and RNA. The purine bases are adenine (A) and guanine (G). The pyrimidine bases are thymine (T) and cytosine (C) in DNA. In RNA, uracil (U) is present instead of thymine. The nucleotide bases are part of a subunit of DNA, the nucleotide. This consists of one of the four nucleotide bases, a sugar (deoxyribose), and a phosphate group. The nitrogen atom in position 9 of a purine or in position 1 of a pyrimidine is bound to the carbon in position 1 of the sugar (*N*-glycosidic bond). Ribonucleic acid (RNA) differs from DNA in two respects: it contains ribose instead of deoxyribose (unlike the latter, ribose has a hydroxyl group on the position 2 carbon atom) and uracil (U) instead of thymine. Uracil does not have a methyl group at position C5.

B. Nucleotide chain

DNA is a polymer of deoxyribonucleotide units. The nucleotide chain is formed by joining a hydroxyl group on the sugar of one nucleotide to the phosphate group attached to the sugar of the next nucleotide. The sugars linked together by the phosphate groups form the invariant part of the DNA. The variable part is in the sequence of the nucleotide bases A, T, C, and G. A DNA nucleotide chain is polar. The polarity results from the way the sugars are attached to each other. The phosphate group at position C5 (the 5' carbon) of one sugar joins to the hydroxyl group at position C3 (the 3' carbon) of

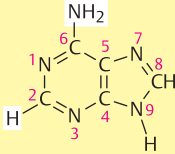
the next sugar by means of a phosphate diester bridge. Thus, one end of the chain has a 5' triphosphate group free and the other end has a 3' hydroxy group free (5' end and 3' end, respectively). By convention, the sequence of nucleotide bases is written in the 5' to 3' direction.

C. Spatial relationship

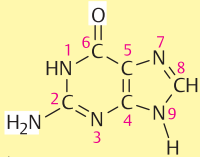
The chemical structure of the nucleotide bases determines a defined spatial relationship. Within the double helix, a purine (adenine or guanine) always lies opposite a pyrimidine (thymine or cytosine). Three hydrogen-bond bridges are formed between cytosine and guanine, and two between thymine and adenine. Therefore, only guanine and cytosine or adenine and thymine can lie opposite and pair with each other (complementary base pairs G – C and A – T). Other spatial relationships are not usually possible.

D. DNA double strand

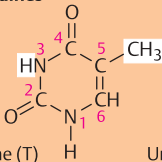
DNA forms a double strand. As a result of the spatial relationships of the nucleotide bases, a cytosine will always lie opposite to a guanine and a thymine to an adenine. The sequence of the nucleotide bases on one strand of DNA (in the 5' to 3' direction) is complementary to the nucleotide base sequence (or simply the base sequence) of the other strand in the 3' to 5' direction. The specificity of base pairing is the most important structural characteristic of DNA.

Purines

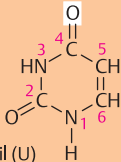
Adenine (A)



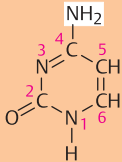
Guanine (G)

Pyrimidines

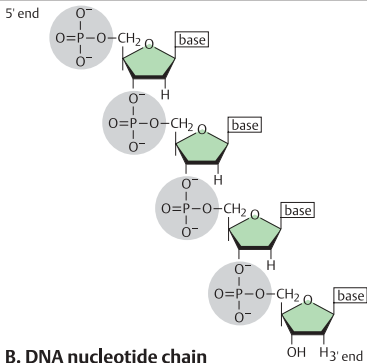
Thymine (T)



Uracil (U)

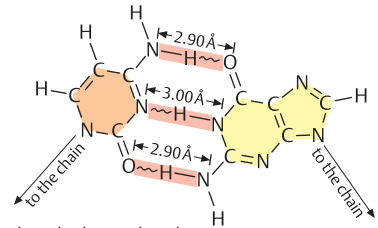


Cytosine (C)

A. Nucleotide bases**B. DNA nucleotide chain**

Cytosine

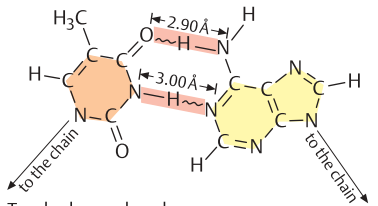
Guanine



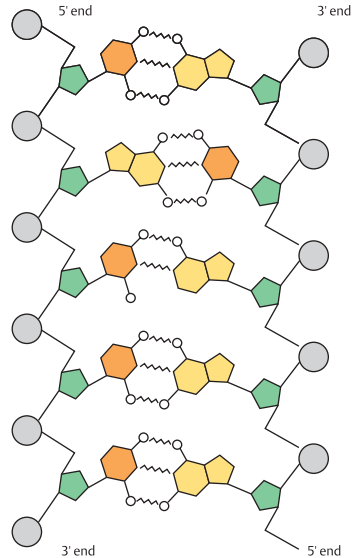
Three hydrogen bonds

Thymine

Adenine



Two hydrogen bonds

C. Spatial relationship**D. Double strand of DNA**

DNA Structure

In 1953, James Watson and Francis Crick recognized that DNA must exist as a double helix. This structure explains both important functional aspects: replication and the transmission of genetic information. The elucidation of the structure of DNA is considered as the beginning of the development of modern genetics. With it, gene structure and function can be understood at the molecular level.

A. DNA as a double helix

The double helix is the characteristic structural feature of DNA. The two helical polynucleotide chains are wound around each other along a common axis. The nucleotide base pairs (bp), either A–T or G–C, lie within. The diameter of the helix is 20 Å (2×10^{-7} mm). Neighboring bases lie 3.4 Å apart. The helical structure repeats itself at intervals of 34 Å, or every ten base pairs. Because of the fixed spatial relationship of the nucleotide bases within the double helix and opposite each other, the two chains of the double helix are exactly complementary. The form illustrated here is the so-called B form (B-DNA). Under certain conditions, DNA can also assume other forms (Z-DNA, A-DNA, see p. 41).

B. Replication

Since the nucleotide chains lying opposite each other within the double helix are strictly complementary, each can serve as a pattern (template) for the formation (replication) of a new chain when the helix is opened. DNA replication is semiconservative, i.e., one completely new strand will be formed and one strand retained.

C. Denaturation and renaturation

The noncovalent hydrogen bonds between the nucleotide base pairs are weak. Nevertheless, DNA is stable at physiological temperatures because it is a very long molecule. The two complementary strands can be separated (denaturation) by means of relatively weak chemical reagents (e.g., alkali, formamide, or urea) or by careful heating. The resulting single-stranded molecules are relatively stable. With cooling, complementary single strands can reunite to form double-stranded molecules (renaturation). Noncomplementary single strands do not unite. This is the basis of an important method of identifying nucleic acids: With a single

strand of defined origin, it can be determined with which other single strand it will bind (hybridize). The hybridization of complementary segments of DNA is an important principle in the analysis of genes.

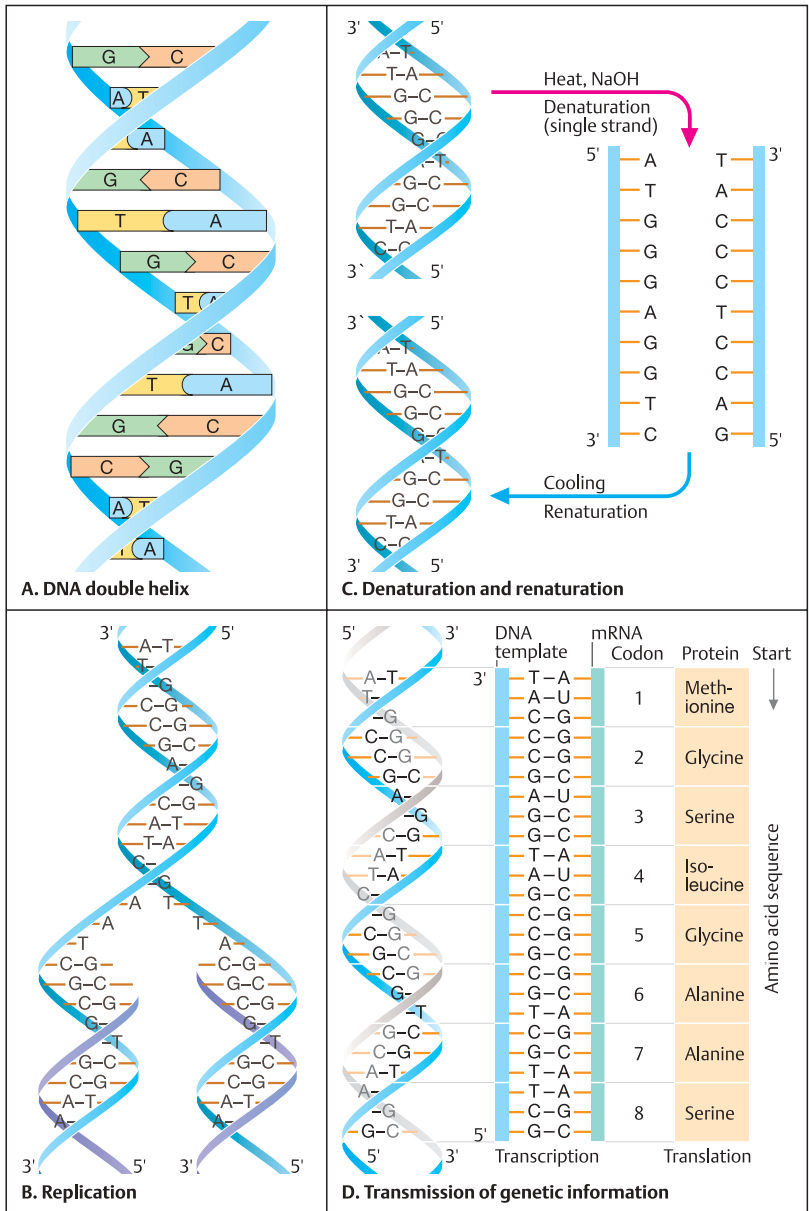
D. Transmission of genetic information

Genetic information lies in the sequence of nucleotide base pairs (A–T or G–C). A sequence of three base pairs represents a code word (codon) for an amino acid. The codon sequence determines a corresponding sequence of amino acids. These form a polypeptide (gene product). The sequence of the nucleotide bases is first transferred (transcription) from one DNA strand to a further information-bearing molecule (mRNA, messenger RNA). Then the nucleotide base sequence of the mRNA serves as a template for a sequence of amino acids corresponding to the order of the codons (translation).

A gene can be defined as a section of DNA responsible for the formation of a polypeptide (one gene, one polypeptide). One or more polypeptides form a protein. Thus, several genes may be involved in the formation of a protein.

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Alternative DNA Structures

Gene expression and transcription can be influenced by changes of DNA topology. However, this type of control of gene expression is relatively universal and nonspecific. Thus, it is more suitable for permanent suppression of transcription, e.g., in genes that are expressed only in certain tissues or are active only during the embryonic period and later become permanently inactive.

A. Three forms of DNA

The DNA double helix does not occur as a single structure, but rather represents a structural family of different types. The original classic form, determined by Watson and Crick in 1953, is B-DNA. The essential structural characteristic of B-DNA is the formation of two grooves, one large (major groove) and one small (minor groove). There are at least two further, alternative forms of the DNA double helix, Z-DNA and the rare form A-DNA. While B-DNA forms a right-handed helix, Z-DNA shows a left-handed conformation. This leads to a greater distance (0.77 nm) between the base pairs than in B-DNA and a zigzag form (thus the designation Z-DNA). A-DNA is rare. It exists only in the dehydrated state and differs from the B form by a 20-degree rotation of the perpendicular axis of the helix. A-DNA has a deep major groove and a flat minor groove (Figures from Watson et al, 1987).

B. Major and minor grooves in B-DNA

The base pairing in DNA (adenine–thymine and guanine–cytosine) leads to the formation of a large and a small groove because the glycosidic bonds to deoxyribose (dRib) are not diametrically opposed. In B-DNA, the purine and pyrimidine rings lie 0.34 nm apart. DNA has ten base pairs per turn of the double helix. The distance from one complete turn to the next is 3.4 nm. In this way, localized curves arise in the double helix. The result is a somewhat larger and a somewhat smaller groove.

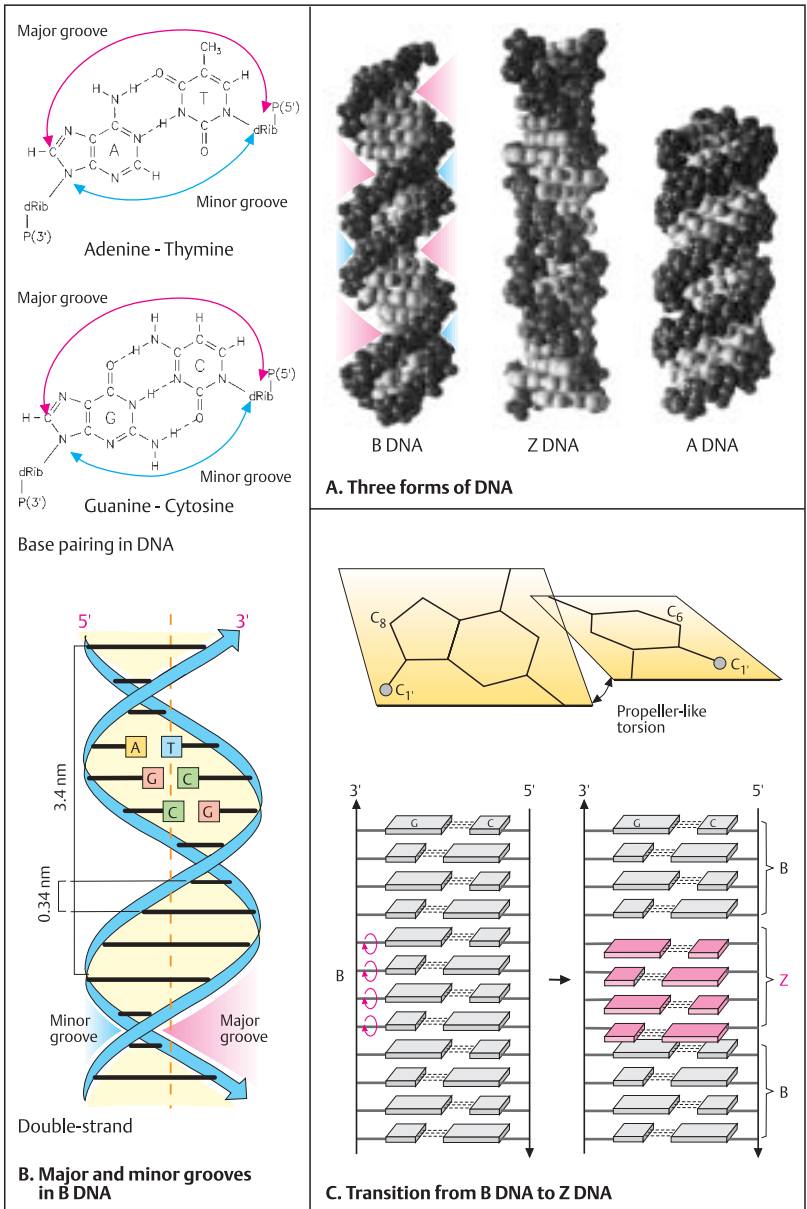
C. Transition from B-DNA to Z-DNA

B-DNA is a perfect regular double helix except that the base pairs opposite each other do not lie exactly at the same level. They are twisted in a propeller-like manner. In this way, DNA can easily be bent without causing essential changes in the local structures.

In Z-DNA the sugar–phosphate skeleton has a zigzag pattern; the single Z-DNA groove has a greater density of negatively charged molecules. Z-DNA may occur in limited segments *in vivo*. A segment of B-DNA consisting of GC pairs can be converted into Z-DNA when the bases are rotated 180 degrees. Normally, Z-DNA is thermodynamically relatively unstable. However, transition to Z-DNA is facilitated when cytosine is methylated in position 5 (C5). The modification of DNA by methylation of cytosine is frequent in certain regions of DNA of eukaryotes. There are specific proteins that bind to Z-DNA, but their significance for the regulation of transcription is not clear.

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DNA Replication

DNA synthesis involves a highly coordinated action of many proteins. Precision and speed are required. The two new DNA chains are assembled at a rate of about 1000 nucleotides per second in *E. coli*. The principal enzymatic proteins are polymerases, which carry out template-directed synthesis; helicases, which separate the two strands to generate the replication fork (see D); primases, which initiate chain synthesis at preferred sites; initiation proteins, which recognize the origin of replication point; and proteins that remodel the double helix. The entire complex is called the replisome. In their paper elucidating the structure of DNA, Watson and Crick (1953) noted in closing, "It has not escaped our attention that this structure immediately suggests a copying mechanism for the genetic material," at that time an unsolved problem. Although biochemically complex, DNA replication is genetically relatively simple. During replication, each strand of DNA serves as a template for the formation of a new strand (semiconservative replication).

A. Prokaryote replication begins at a single site

In prokaryote cells, replication begins at a defined point in the ring-shaped bacterial chromosome, the origin of replication (1). From here, new DNA is formed at the same speed in both directions until the DNA has been completely duplicated and two chromosomes are formed. Replication can be visualized by autoradiography after the newly replicated DNA has incorporated tritium (^3H)-labeled thymidine (2).

B. Eukaryote replication begins at several sites

DNA synthesis occurs during a defined phase of the cell cycle (S phase). This would take a very long time if there were only one starting point. However, replication of eukaryotic DNA begins at numerous sites (replicons) (1). It proceeds in both directions from each replicon until neighboring replicons fuse (2) and all of the DNA is duplicated (3). The electron micrograph (4) shows replicons at three sites.

C. Scheme of replication

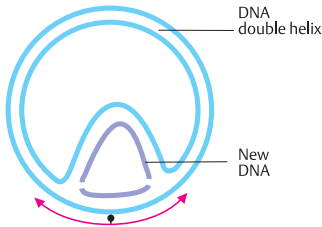
New DNA is synthesized in the 5' to 3' direction, but not in the 3' to 5' direction. A new nucleotide cannot be attached to the 5'-OH end of the new nucleotide chain. Only at the 3' end can nucleotides be attached continuously. New DNA at the 5' end is replicated in small segments. This represents an obstacle at the end of a chromosome (telomere, see p. 180).

D. Replication fork

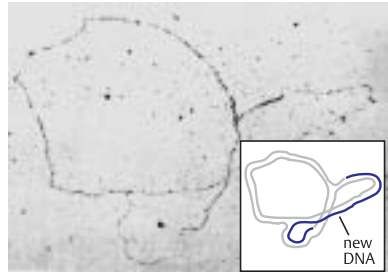
At the replication fork, each of the two DNA strands serves as a template for the synthesis of new DNA. First, the double helix at the replication fork region is unwound by an enzyme system (topoisomerases). Since the parent strands are antiparallel, DNA replication can proceed continuously in only one DNA strand (5' to 3' direction) (leading strand). Along the 3' to 5' strand (lagging strand), the new DNA is formed in small segments of 1000–2000 bases (Okazaki fragments). In this strand a short piece of RNA is required as a primer to start replication. This is formed by an RNA polymerase (primase). The RNA primer is subsequently removed; DNA is inserted into the gap by polymerase I and, finally, the DNA fragments are linked by DNA ligase. The enzyme responsible for DNA synthesis (DNA polymerase III) is complex and comprises several subunits. There are different enzymes for the leading and lagging strands in eukaryotes. During replication, mistakes are eliminated by a complex proof-reading mechanism that removes any incorrectly incorporated bases and replaces them with the correct ones.

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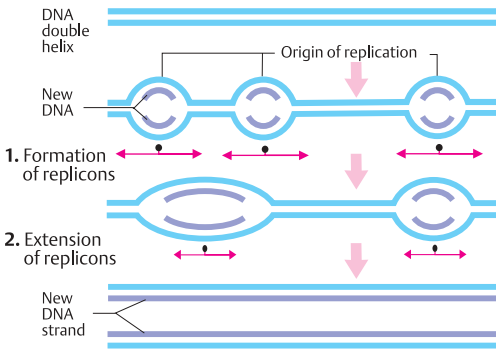


1. DNA replication in the bacterial chromosome



2. Prokaryotic replication in an autoradiogram in *E. coli* (J. Cairns)

A. Prokaryotic replication begins at one site

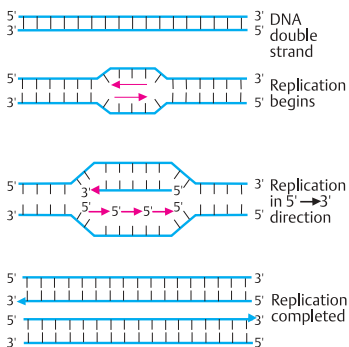


3. Replication completed

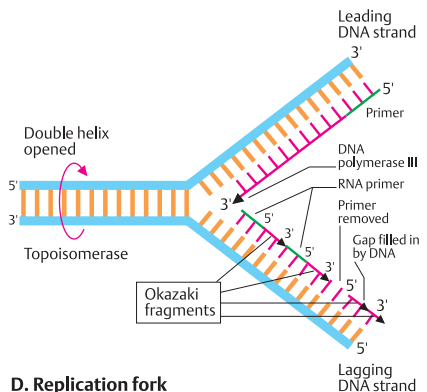


4. Eukaryotic replication in the EM (D. S. Hogness)

B. Eukaryotic replication begins at several sites



C. Scheme of replication



D. Replication fork

The Flow of Genetic Information: Transcription and Translation

The information contained in the nucleotide sequence of a gene must be converted into useful biological function. This is accomplished by proteins, either directly, by being involved in a biochemical pathway, or indirectly, by regulating the activity of a gene. The flow of genetic information is unidirectional and requires two major steps: *transcription* and *translation*. First, the information of the coding sequences of a gene is transcribed into an intermediary RNA molecule, which is synthesized in sequences that are precisely complementary to those of the coding strand of DNA (transcription). During the second major step the sequence information in the messenger RNA molecule (mRNA) is translated into a corresponding sequence of amino acids (translation). The length and sequence of the amino acid chain specified by a gene results in a polypeptide with a biological function (gene product).

A. Transcription

First, the nucleotide sequence of one strand of DNA is transcribed into a complementary molecule of RNA (messenger RNA, mRNA). The DNA helix is opened by a complex set of proteins. The DNA strand in the 3' to 5' direction (coding strand) serves as the template for the transcription into RNA, which is synthesized in the 5' to 3' direction. It is called the RNA sense strand. RNA transcribed under experimental conditions from the opposing DNA strand is called antisense RNA.

B. Translation

During translation the sequence of codons made up of the nucleotide bases in mRNA is converted into a corresponding sequence of amino acids. Translation occurs in a reading frame which is defined at the start of translation (start codon). Amino acids are joined in the sequence determined by the mRNA nucleotide bases by a further class of RNA, transfer RNA (tRNA). Each amino acid has its own tRNA, which has a region that is complementary to its codon of the mRNA (anticodon). The codons 1, 2, 3, and 4 of the mRNA are translated into the amino acid sequence methionine (Met), glycine

(Gly), serine (Ser), and isoleucine (Ile), etc. Codon 1 is always AUG (start codon).

C. Stages of translation

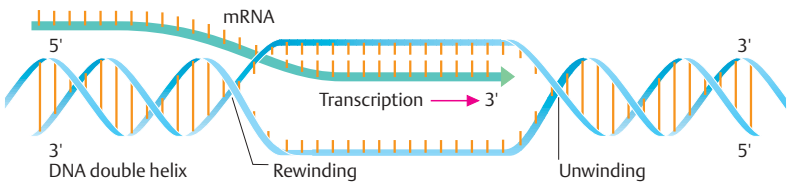
Translation (protein synthesis) in eukaryotes occurs outside of the cell nucleus in ribosomes in the cytoplasm. Ribosomes consist of subunits of numerous associated proteins and RNA molecules (ribosomal RNA, rRNA; p. 204). Translation begins with initiation (1): an initiation complex comprising mRNA, a ribosome, and tRNA is formed. This requires a number of initiation factors (IF1, IF2, IF3, etc.). Then elongation (2) follows: a further amino acid, determined by the next codon, is attached. A three-phase elongation cycle develops, with codon recognition, peptide binding to the next amino acid residue, and movement (translocation) of the ribosome three nucleotides further in the 3' direction of the mRNA. Translation ends with termination (3), when one of three mRNA stop codons (UAA, UGA, or UAG) is reached. The polypeptide chain formed leaves the ribosome, which dissociates into its subunits. The biochemical processes of the stages shown here have been greatly simplified.

D. Structure of transfer RNA (tRNA)

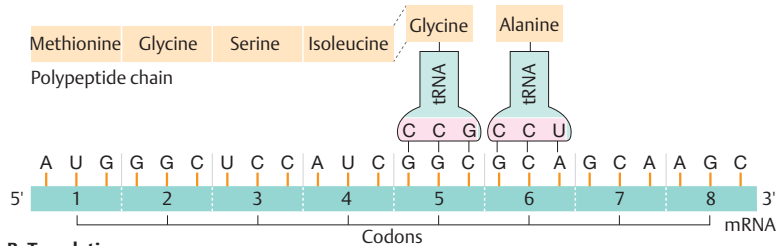
Transfer RNA has a characteristic, cloverleaf-like structure, illustrated here by yeast phenylalanine tRNA (1). It has three single-stranded loop regions and four double-stranded "stem" regions. The three-dimensional structure (2) is complex, but various functional areas can be differentiated, such as the recognition site (anticodon) for the mRNA codon and the binding site for the respective amino acid (acceptor stem) on the 3' end (acceptor end).

References

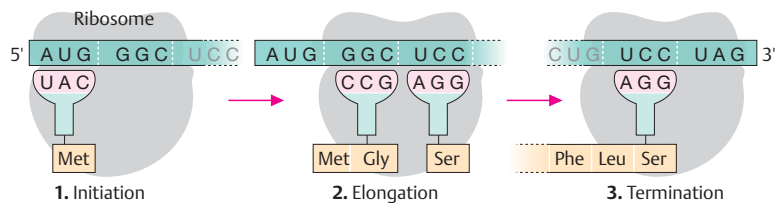
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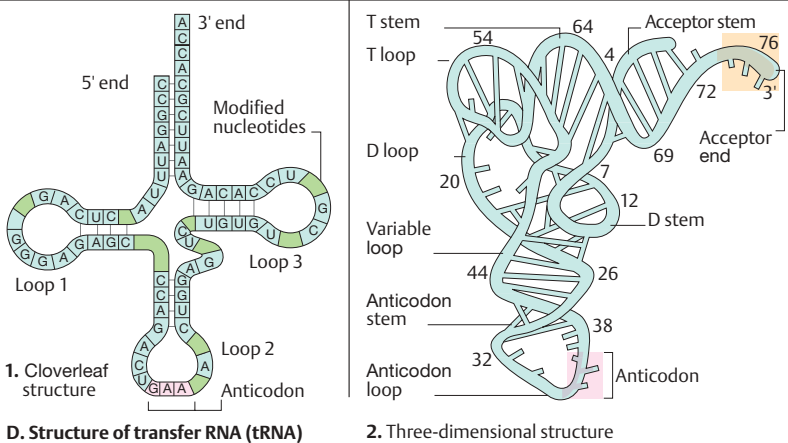
A. Transcription



B. Translation



C. Stages of translation



D. Structure of transfer RNA (tRNA)

Genes and Mutation

The double helix structure of DNA is the basis of both replication and transcription as seen in the preceding pages. The information transmitted during replication and transcription is arranged in units called *genes*. The term gene was introduced in 1909 by the Danish biologist Wilhelm Johannsen (along with the terms genotype and phenotype). Until it was realized that a gene consists of DNA, it was defined in somewhat abstract terms as a factor (Mendel's term) that confers certain heritable properties to a plant or an animal. However, it was not apparent how mutations could be related to the structure of a gene. The discovery that mutations also occur in bacteria and other microorganisms paved the way to understanding their nature (see p. 84). The organization of genes differs in prokaryotes and eukaryotes as shown below.

A. Transcription in prokaryotes and eukaryotes

Transcription differs in unicellular organisms without a nucleus, such as bacteria (prokaryotes, 1), and in multicellular organisms (eukaryotes, 2), which have a cell nucleus. In prokaryotes, the mRNA serves directly as a template for translation. The sequences of DNA and mRNA correspond in a strict 1:1 relationship, i.e., they are colinear. Translation begins even before transcription has completely ended. In contrast, a primary transcript of RNA precursor mRNA) is formed first in eukaryotic cells. This is a preliminary form of the mature mRNA. The mature mRNA is formed when the noncoding sections are removed from the primary transcript, before it leaves the nucleus to act as a template for forming a polypeptide (RNA processing).

The reason for these important differences is that functionally related genes generally lie together in prokaryotes and that noncoding segments (introns) are present in the genes of eukaryotes (see p. 50).

B. DNA and mutation

Coding DNA and its corresponding polypeptide are colinear. An alteration (mutation) of the DNA base sequence may lead to a different codon. The position of the resulting change in the sequence of amino acids corresponds to the position of the mutation (1). Panel B shows the

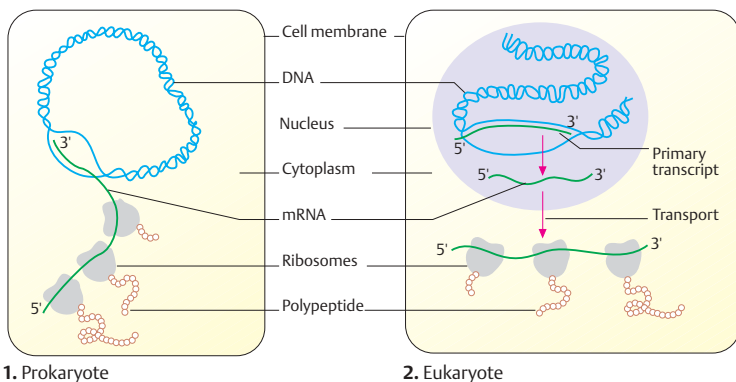
gene for the protein tryptophan synthetase A of *E. coli* bacteria and mutations at four positions. At position 22, phenylalanine (Phe) has been replaced by leucine (Leu); at position 49, glutamic acid (Glu) by glutamine (Gln); at position 177, Leu by arginine (Arg). Every mutation has a defined position. Whether it leads to incorporation of another amino acid depends on how the corresponding codon has been altered. Different mutations at one position (one codon) in different DNA molecules are possible (2). Two different mutations have been observed at position 211: glycine (Gly) to arginine (Arg) and Gly to glutamic acid (Glu). Normally (in the wild-type), codon 211 is GGA and codes for glycine (3). A mutation of GGA to AGA leads to a codon for arginine; a mutation to GAA leads to a codon for glutamic acid (4).

C. Types of mutation

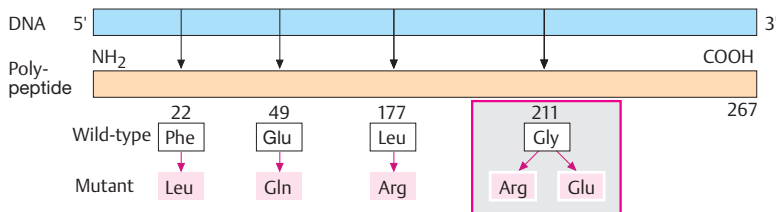
Basically, there are three different types of mutation involving single nucleotides (point mutation): substitution (exchange), deletion (loss), and insertion (addition). With substitution, the consequences depend on how a codon has been altered. Two types of substitution are distinguished: transposition (exchange of one purine for another purine or of one pyrimidine for another) and transversion (exchange of a purine for a pyrimidine, or vice versa). A substitution may alter a codon so that a wrong amino acid is present at this site but has no effect on the reading frame (missense mutation), whereas a deletion or insertion causes a shift of the reading frame (frameshift mutation). Thus the sequences that follow no longer code for a functional gene product (nonsense mutation).

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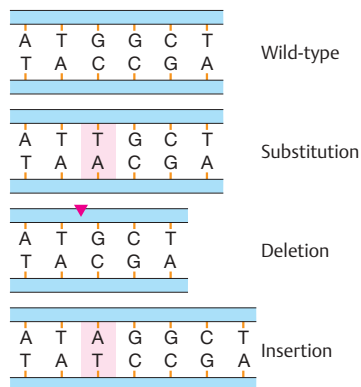
A. Transcription and translation in prokaryotes and eukaryotes



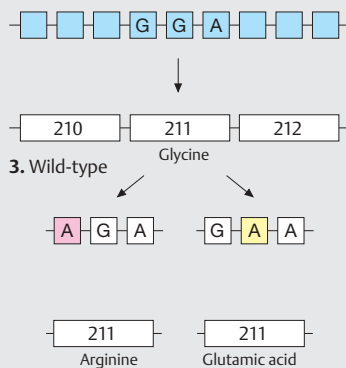
1. Defined position of a mutation

B. DNA and mutation

2. Different mutations of one codon



C. Types of mutation



4. Different mutations

Genetic Code

The genetic code is the set of biological rules by which DNA nucleotide base pair sequences are translated into corresponding sequences of amino acids. Genes do not code for proteins directly, but do so through a messenger molecule (messenger RNA, mRNA). A code word (codon) for an amino acid consists of a sequence of three nucleotide base pairs (triplet codon). The genetic code also includes sequences for the beginning (start codon) and for the end (stop codon) of the coding region. The genetic code is universal; the same codons are used by different organisms.

A. Genetic code in mRNA for all amino acids

Each codon corresponds to one amino acid, but one amino acid may be coded for by different codons (redundancy of the code). For example, there are two possibilities to code for the amino acid phenylalanine: UUU and UUC, and there are six possibilities to code for the amino acid serine: UCU, UCC, UCA, UCG, AGU, and AGC. Many amino acids are determined by more than one codon. The greatest variation is in the third position (at the 3' end of the triplet). The genetic code was elucidated in 1966 by analyzing how triplets transmit information from the genes to proteins. mRNA added to bacteria could be directly converted into a corresponding protein. Synthetic RNA polymers such as polyuridylylate (poly (U)), polyadenylate (poly(A)), and polycytidylylate (poly(C)) could be directly translated into polyphenylalanine, polylysine, and polyproline in extracts of *E. coli* bacteria. This showed that UUU must code for phenylalanine, AAA for lysine, and CCC for proline. By further experiments with mixed polymers of different proportions of two or three nucleotides, the genetic code was determined for all amino acids and all nucleotide compositions.

B. Abbreviated code

Sequences of amino acids are designated with the single-letter abbreviations ("alphabetic code").

The start codon is AUG (methionine). Stop codons are UAA, UAG, and UGA. The only amino acids that are encoded by a single codon are methionine (AUG) and tryptophan (UGG).

C. Open reading frame (ORF)

A segment of a nucleotide sequence can correspond to one of three reading frames (e.g., A, B, or C); however, only one is correct (open reading frame). In the example shown, the reading frames B and C are interrupted by a stop codon after three and five codons, respectively. Thus they cannot serve as reading frames for a coding sequence. On the other hand, A must be the correct reading frame: It begins with the start codon AUG and yields a sequence without stop codons (open reading frame).

D. Coding by several different nucleotide sequences

Since the genetic code has redundancy, it is possible that different nucleotide sequences code for the same amino acid sequence. However, the differences are limited to one (or at most two) positions of a given triplet codon.

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Nucleotide base					
First	Second				Third
	Uracil (U)	Cytosine (C)	Adenine (A)	Guanine (G)	
Uracil (U)	F Phenylalanine (Phe)	S Serine (Ser)	Y Tyrosine (Tyr)	C Cysteine (Cys)	U
	F Phenylalanine (Phe)	S Serine (Ser)	Y Tyrosine (Tyr)	C Cysteine (Cys)	C
	L Leucine (Leu)	S Serine (Ser)	Stop Codon	Stop Codon	A
	L Leucine (Leu)	S Serine (Ser)	Stop Codon	W Tryptophan (Trp)	G
Cytosine (C)	L Leucine (Leu)	P Proline (Pro)	H Histidine (His)	R Arginine (Arg)	U
	L Leucine (Leu)	P Proline (Pro)	H Histidine (His)	R Arginine (Arg)	C
	L Leucine (Leu)	P Proline (Pro)	Q Glutamine (Gln)	R Arginine (Arg)	A
	L Leucine (Leu)	P Proline (Pro)	Q Glutamine (Gln)	R Arginine (Arg)	G
Adenine (A)	I Isoleucine (Ile)	T Threonine (Thr)	N Asparagine (Asn)	S Serine (Ser)	U
	I Isoleucine (Ile)	T Threonine (Thr)	N Asparagine (Asn)	S Serine (Ser)	C
	I Isoleucine (Ile)	T Threonine (Thr)	K Lysine (Lys)	R Arginine (Arg)	A
	Start (Methionine)	T Threonine (Thr)	K Lysine (Lys)	R Arginine (Arg)	G
Guanine (G)	V Valine (Val)	A Alanine (Ala)	D Aspartic acid (Asp)	G Glycine (Gly)	U
	V Valine (Val)	A Alanine (Ala)	D Aspartic acid (Asp)	G Glycine (Gly)	C
	V Valine (Val)	A Alanine (Ala)	E Glutamic acid (Glu)	G Glycine (Gly)	A
	V Valine (Val)	A Alanine (Ala)	E Glutamic acid (Glu)	G Glycine (Gly)	G

A. Genetic code for all amino acids in mRNA

Start AUG	F (Phe) UUU UUC	L (Leu) CUU CUC CUG CUA UUG UUA	R (Arg) CGU CGC CGG CAA AGA	V GUU GUC GUG GUA
Stop UAA UAG UGA	G (Gly) GGU GGC GGG GGA	M (Met) AUG	S (Ser) UCU UCC UCA AGU AGC	W (Trp) UGG
A (Ala) GCU GCC GCG GCA	H (His) CAU CAC	N (Asn) AAU AAC	T (Thr) ACU ACC ACG ACA	Y (Tyr) UAU UAC
C (Cys) UGU UGC	I (Ile) AUU AUC AUA	P (Pro) CCU CCC CCG CCA		B (Asx) Asn or Asp
D (Asp) GAU GAC		Q (Gln) CAG CAA		Z (Glx) Gln or Glu
E (Glu) GAG GAA	K (Lys) AAG AAA			

B. Abbreviated code

CAGUCU AUGGCAAAUAGGUA GACCAU

A → Met Ala Asn Lys Val Asp His

B → Tyr Gly Lys stop

C → Leu Trp Glu Ile Arg stop

C. Open reading frame (ORF)

UUA CCU AUU AAA AAA CCG

Leu Pro Ile Lys Lys Arg

CUG CCG AUA AAG AAG CGA

D. Coding by several different nucleotide sequences

The Structure of Eukaryotic Genes

Eukaryotic genes consist of coding and noncoding segments of DNA, called exons and introns, respectively. At first glance it seems to be an unnecessary burden to carry DNA without obvious functions within a gene. However, it has been recognized that this has great evolutionary advantages. When parts of different genes are rearranged on new chromosomal sites during evolution, new genes may be constructed from parts of previously existing genes.

A. Exons and introns

In 1977, it was unexpectedly found that the DNA of a eukaryotic gene is longer than its corresponding mRNA. The reason is that certain sections of the initially formed primary RNA transcript are removed before translation occurs. Electron micrographs show that DNA and its corresponding transcript (RNA) are of different lengths (1). When mRNA and its complementary single-stranded DNA are hybridized, loops of single-stranded DNA arise because mRNA hybridizes only with certain sections of the single-stranded DNA. In (2), seven loops (A to G) and eight hybridizing sections are shown (1 to 7 and the leading section L). Of the total 7700 DNA base pairs of this gene (3), only 1825 hybridize with mRNA. A hybridizing segment is called an exon. An initially transcribed DNA section that is subsequently removed from the primary transcript is an intron. The size and arrangement of exons and introns are characteristic for every eukaryotic gene (exon/intron structure). (Electron micrograph from Watson et al., 1987).

B. Intervening DNA sequences (introns)

In prokaryotes, DNA is colinear with mRNA and contains no introns (1). In eukaryotes, mature mRNA is complementary to only certain sections of DNA because the latter contains introns (2). (Figure adapted from Stryer, 1995).

C. Basic eukaryotic gene structure

Exons and introns are numbered in the 5' to 3' direction of the coding strand. Both exons and introns are transcribed into a precursor RNA (primary transcript). The first and the last exons usually contain sequences that are not trans-

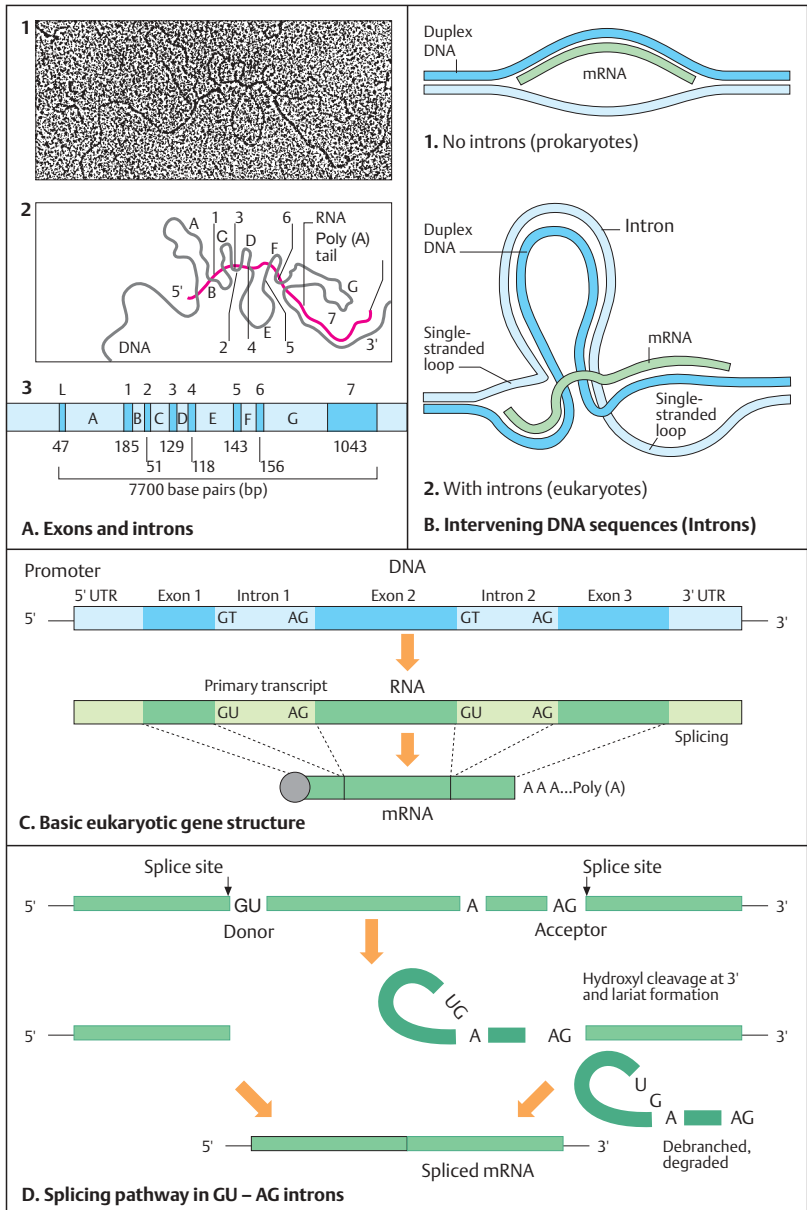
lated. These are called the 5' untranslated region (5' UTR) of exon 1 and the 3' UTR at the 3' end of the last exon. The noncoding segments (introns) are removed from the primary transcript and the exons on either side are connected by a process called splicing. Splicing must be very precise to avoid an undesirable change of the correct reading frame. Introns almost always start with the nucleotides GT in the 5' to 3' strand (GU in RNA) and end with AG. The sequences at the 5' end of the intron beginning with GT are called splice donor site and at the 3' end, ending with AG, are called the splice acceptor site. Mature mRNA is modified at the 5' end by adding a stabilizing structure called a "cap" and by adding many adenines at the 3' end (polyadenylation) (see p. 50).

D. Splicing pathway in GU-AG introns

RNA splicing is a complex process mediated by a large RNA-containing protein called a spliceosome. This consists of five types of small nuclear RNA molecules (snRNA) and more than 50 proteins (small nuclear riboprotein particles). The basic mechanism of splicing schematically involves autocatalytic cleavage at the 5' end of the intron resulting in lariat formation. This is an intermediate circular structure formed by connecting the 5' terminus (UG) to a base (A) within the intron. This site is called the branch site. In the next stage, cleavage at the 3' site releases the intron in lariat form. At the same time the right exon is ligated (spliced) to the left exon. The lariat is debranched to yield a linear intron and this is rapidly degraded. The branch site identifies the 3' end for precise cleavage at the splice acceptor site. It lies 18–40 nucleotides upstream (in 5' direction) of the 3' splice site. (Figure adapted from Strachan and Read, 1999).

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DNA Sequencing

Knowledge of the nucleotide sequence of a gene provides important information about its structure, function, and evolutionary relationship to other similar genes in the same or different organisms. Thus, the development in the 1970s of relatively simple methods for sequencing DNA has had a great impact on genetics. Two basic methods for DNA sequencing have been developed: a chemical cleavage method (A. M. Maxam and W. Gilbert, 1977) and an enzymatic method (F. Sanger, 1981). A brief outline of the underlying principles follows.

A. Sequencing by chemical degradation

This method utilizes base-specific cleavage of DNA by certain chemicals. Four different chemicals are used in four reactions, one for each base. Each reaction produces a set of DNA fragments of different sizes. The sizes of the fragments in a reaction mixture are determined by positions in the DNA of the nucleotide that has been cleaved. A double-stranded or single-stranded fragment of DNA to be sequenced is processed to obtain a single strand labeled with a radioactive isotope at the 5' end (1). This DNA strand is treated with one of the four chemicals for one of the four reactions. Here the reaction at guanine sites (G) by dimethyl sulfate (DMS) is shown. Dimethyl sulfate attaches a methyl group to the purine ring of G nucleotides. The amount of DMS used is limited so that on average just one G nucleotide per strand is methylated, not the others (shown here in four different positions of G). When a second chemical, piperidine, is added, the nucleotide purine ring is removed and the DNA molecule is cleaved at the phosphodiester bond just upstream of the site without the base. The overall procedure results in a set of labeled fragments of defined sizes according to the positions of G in the DNA sample being sequenced. Similar reactions are carried out for the other three bases (A, T, and C, not shown). The four reaction mixtures, one for each of the bases, are run in separate lanes of a polyacrylamide gel electrophoresis. Each of the four lanes represents one of the four bases G, A, T, or C. The smallest fragment will migrate the farthest downward, the next a little less far, etc. One can then read the sequence in the direction opposite to migration to obtain the sequence in

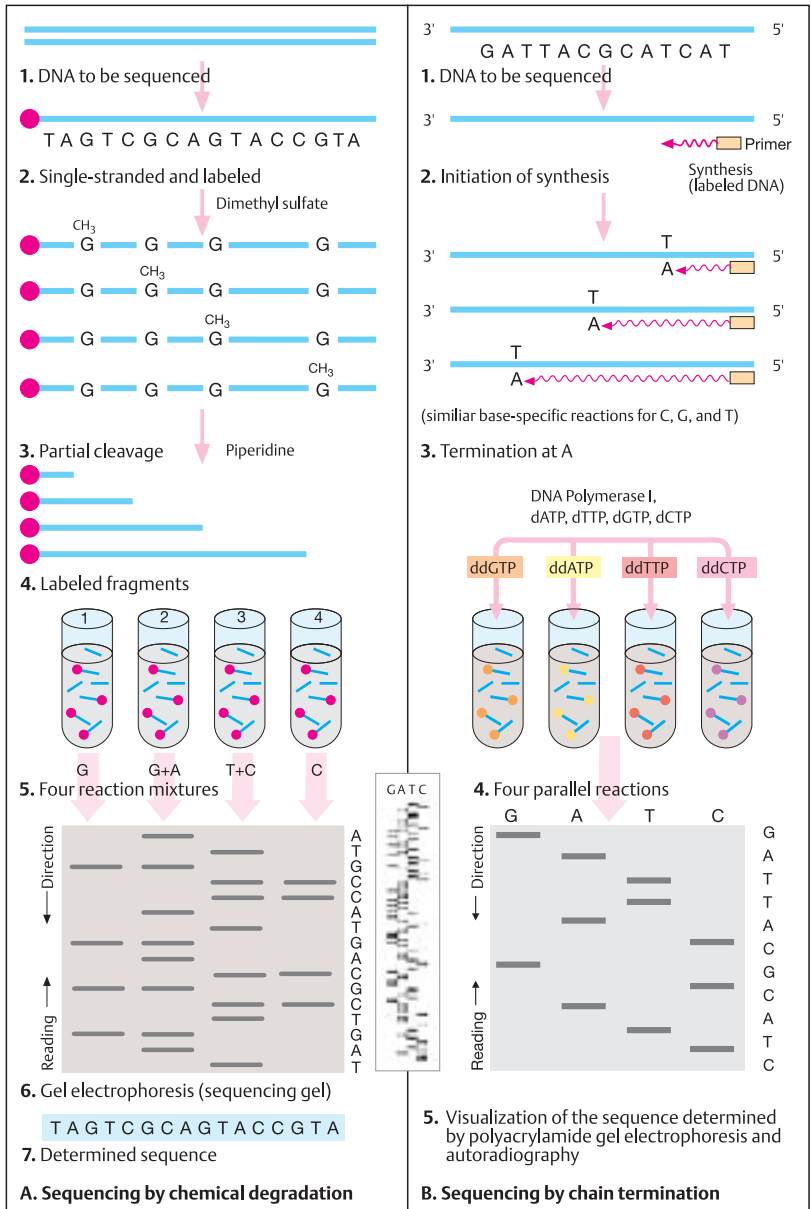
the 5' to 3' direction (here TAGTCGCAG-TACCGTA).

B. Sequencing by chain termination

This method, now much more widely used than the chemical cleavage method, rests on the principle that DNA synthesis is terminated when instead of a normal deoxynucleotide (dATP, dTTP, dGTP, dCTP), a dideoxynucleotide (ddATP, ddTTP, ddGTP, ddCTP) is used. A dideoxynucleotide (ddNTP) is an analogue of the normal dNTP. It differs by lack of a hydroxyl group at the 3' carbon position. When a dideoxynucleotide is incorporated during DNA synthesis, no bond between its 3' position and the next nucleotide is possible because the ddNTP lacks the 3' hydroxyl group. Thus, synthesis of the new chain is terminated at this site. The DNA fragment to be sequenced has to be single-stranded (1). DNA synthesis is initiated using a primer and one of the four ddNTPs labeled with ^{32}P in the phosphate groups or, for automated sequencing, with a fluorophore (see next plate). Here an example of chain termination using ddATP is shown (3). Wherever an adenine (A) occurs in the sequence, the dideoxyadenine triphosphate will cause termination of the new DNA chain being synthesized. This will produce a set of different DNA fragments whose sizes are determined by the positions of the adenine residues occurring in the fragment to be sequenced. Similar reactions are done for the other three nucleotides. The four parallel reactions will yield a set of fragments with defined sizes according to the positions of the nucleotides where the new DNA synthesis has been terminated. The fragments are separated according to size by gel electrophoresis as in the chemical method. The sequence gel is read in the direction from small fragments to large fragments to derive the nucleotide sequence in the 5' to 3' direction. An example of an actual sequencing gel is shown between panel A and B.

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Automated DNA Sequencing

Large-scale DNA sequencing requires automated procedures based on fluorescence labeling of DNA and suitable detection systems. In general, a fluorescent label can be used either directly or indirectly. Direct fluorescent labels, as used in automated sequencing, are fluorophores. These are molecules that emit a distinct fluorescent color when exposed to UV light of a specific wavelength. Examples of fluorophores used in sequencing are fluorescein, which fluoresces pale green when exposed to a wavelength of 494 nm; rhodamine, which fluoresces red at 555 nm; and aminomethylcumarin acetic acid, which fluoresces blue at 399 nm. In addition, a combination of different fluorophores can be used to produce a fourth color. Thus, each of the four bases can be distinctly labeled.

Another approach is to use PCR-amplified products (thermal cycle sequencing, see A). This has the advantage that double-stranded rather than single-stranded DNA can be used as the starting material. And since small amounts of template DNA are sufficient, the DNA to be sequenced does not have to be cloned beforehand.

A. Thermal cycle sequencing

The DNA to be sequenced is contained in vector DNA (1). The primer, a short oligonucleotide with a sequence complementary to the site of attachment on the single-stranded DNA, is used as a starting point. For sequencing short stretches of DNA, a universal primer is sufficient. This is an oligonucleotide that will bind to vector DNA adjacent to the DNA to be sequenced. However, if the latter is longer than about 750 bp, only part of it will be sequenced. Therefore, additional internal primers are required. These anneal to different sites and amplify the DNA in a series of contiguous, overlapping chain termination experiments (2). Here, each primer determines which region of the template DNA is being sequenced.

In thermal cycle sequencing (3), only one primer is used to carry out PCR reactions, each with one dideoxynucleotide (ddA, ddT, ddG, or ddC) in the reaction mixture. This generates a series of different chain-terminated strands, each dependent on the position of the particular nucleotide base where the chain is being terminated (4). After many cycles and with elec-

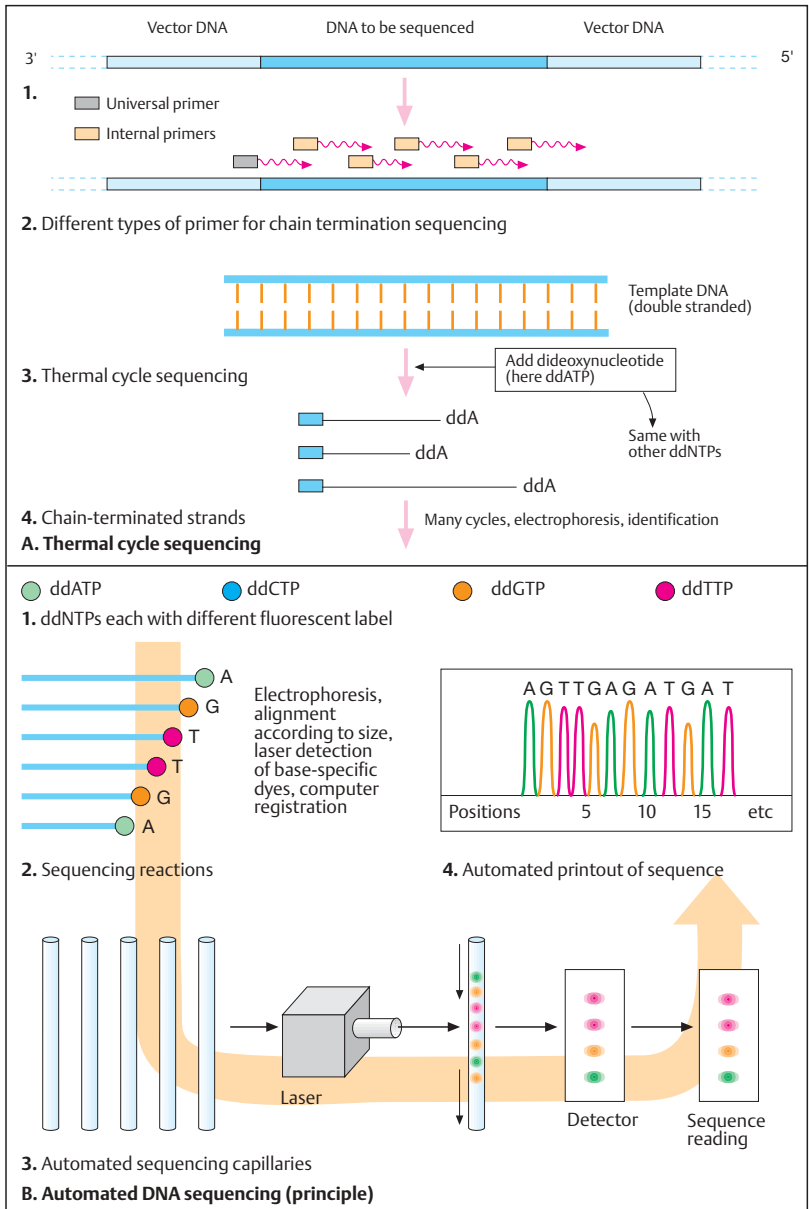
trophoresis, the sequence can be read as shown in the previous plate. One advantage of thermal cycle sequencing is that double-stranded DNA can be used as starting material. (Illustration based on Figures 4.5 and 4.6 in Brown, 1999).

B. Automated DNA sequencing (principle)

Automated DNA sequencing involves four fluorophores, one for each of the four nucleotide bases. The resulting fluorescent signal is recorded at a fixed point when DNA passes through a capillary containing an electrophoretic gel. The base-specific fluorescent labels are attached to appropriate dideoxynucleotide triphosphates (ddNTP). Each ddNTP is labeled with a different color, e.g., ddATP green, ddCTP blue, ddGTP yellow, and ddTTP red (1). (The actual colors for each nucleotide may be different.) All chains terminated at an adenine (A) will yield a green signal; all chains terminated at a cytosine (C) will yield a blue signal, and so on. The sequencing reactions based on this kind of chain termination at labeled nucleotides (2) are carried out automatically in sequencing capillaries (3). The electrophoretic migration of the ddNTP-labeled chains in the gel in the capillary pass in front of a laser beam focused on a fixed position. The laser induces a fluorescent signal that is dependent on the specific label representing one of the four nucleotides. The sequence is electronically read and recorded and is visualized as alternating peaks in one of the four colors, representing the alternating nucleotides in their sequence positions. In practice the peaks do not necessarily show the same maximal intensity as in the schematic diagram shown here. (Illustration based on Brown, 1999, and Strachan and Read, 1999).

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DNA Cloning

To obtain sufficient amounts of a specific DNA sequence (e.g., a gene of interest) for study, it must be selectively amplified. This is accomplished by DNA cloning, which produces a homogeneous population of DNA fragments from a mixture of very different DNA molecules or from all the DNA of the genome. Here procedures are required to identify DNA from the correct region in the genome, to separate it from other DNA, and to multiply (clone) it selectively. Identification of the correct DNA fragment utilizes the specific hybridization of complementary single-stranded DNA (molecular hybridization). A short segment of single-stranded DNA, a probe, originating from the sequence to be studied, will hybridize to its complementary sequences after these have been denatured (made single-stranded, see Southern blot analysis, p. 62). After the hybridized sequence has been separated from other DNA, it can be cloned. The selected DNA sequences can be amplified in two basic ways: in cells (cell-based cloning) or by cell-free cloning (see polymerase chain reaction, PCR, p. 66).

A. Cell-based DNA cloning

Cell-based DNA cloning requires four initial steps. First, a collection of different DNA fragments (here labeled 1, 2, and 3) are obtained from the desired DNA (target DNA) by cleaving it with a restriction enzyme (see p. 64). Since fragments resulting from restriction enzyme cleavage have a short single-stranded end of a specific sequence at both ends, they can be ligated to other DNA fragments that have been cleaved with the same enzyme. The fragments produced in step 1 are joined to DNA fragments containing the origin of replication (OR) of a replicon, which enables them to replicate (2). In addition, a fragment may be joined to a selectable marker, e.g., a DNA sequence containing an antibiotic resistance gene. The recombinant DNA molecules are transferred into host cells (bacterial or yeast cells). Here the recombinant DNA molecules can replicate independently of the host cell genome (3). Usually the host cell takes up only one (although occasionally more than one) foreign DNA molecule. The host cells transformed by recombinant (foreign) DNA are grown in culture and multiplied (propagation, 4). Selective growth of one of the

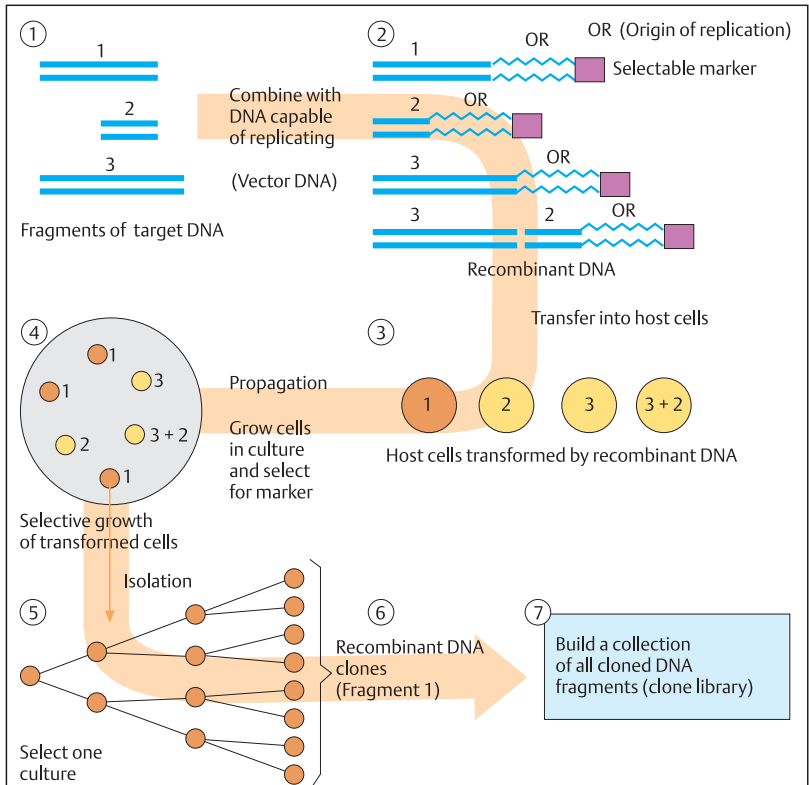
cell clones allows isolation of one type of recombinant DNA molecule (5). After further propagation, a homogeneous population of recombinant DNA molecules is obtained (6). A collection of different fragments of cloned DNA is called a clone library (7, see DNA libraries). In cell-based cloning, the replicon-containing DNA molecules are referred to as vector molecules. (Figure adapted from Strachan and Read, 1999).

B. A plasmid vector for cloning

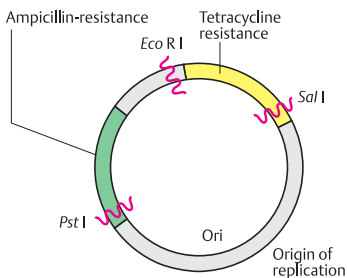
Many different vector systems exist for cloning DNA fragments of different sizes. Plasmid vectors are used to clone small fragments. The experiment is designed in such a way that incorporation of the fragment to be cloned changes the plasmid's antibiotic resistance to allow selection for these recombinant plasmids. A formerly frequently used plasma vector (pBR322) is presented. This plasmid contains recognition sites for the restriction enzymes *Pst*I, *Eco*RI, and *Sal*I in addition to genes for ampicillin and tetracycline resistance (1). If a foreign DNA fragment is incorporated into the plasmid at the site of the *Eco*RI recognition sequence, then tetracycline and ampicillin resistance will be retained (2). If the enzyme *Pst*I is used to incorporate the fragment to be used, ampicillin resistance is lost (the bacterium becomes ampicillin sensitive), but tetracycline resistance is retained. If the enzyme *Sal*I is used to incorporate the fragment, tetracycline resistance disappears (the bacterium becomes tetracycline sensitive), but ampicillin resistance is retained. Thus, depending on how the fragment has been incorporated, recombinant plasmids containing the DNA fragment to be cloned can be distinguished from nonrecombinant plasmids by altered antibiotic resistance. Cloning in plasmids (bacteria) has become less important since yeast artificial chromosomes (YACs) have become available for cloning relatively large DNA fragments (see p. 110).

References

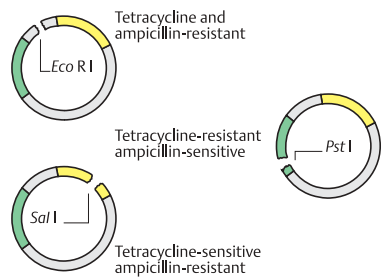
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A. Cell-based DNA cloning



1. Genes for antibiotic resistance



2. Altered antibiotic resistance

B. A plasmid vector (pBR 322) for cloning

cDNA Cloning

cDNA is a single-stranded segment of DNA that is complementary to the mRNA of a coding DNA segment or of a whole gene. It can be used as a probe (cDNA probe as opposed to a genomic probe) for the corresponding gene because it is complementary to coding sections (exons) of the gene. If the gene has been altered by structural rearrangement at a corresponding site, e.g., by deletion, the normal and mutated DNA can be differentiated. Thus, the preparation and cloning of cDNA is of great importance. From the cDNA sequence, essential inferences can be made about a gene and its gene product.

A. Preparation of cDNA

cDNA is prepared from mRNA. Therefore, a tissue is required in which the respective gene is transcribed and mRNA is produced in sufficient quantities. First, mRNA is isolated. Then a primer is attached so that the enzyme reverse transcriptase can form complementary DNA (cDNA) from the mRNA. Since mRNA contains poly(A) at its 3' end, a primer of poly(T) can be attached. From here, the enzyme reverse transcriptase can start forming cDNA in the 5' to 3' direction. The RNA is then removed by ribonuclease. The cDNA serves as a template for the formation of a new strand of DNA. This requires the enzyme DNA polymerase. The result is a double strand of DNA, one strand of which is complementary to the original mRNA. To this DNA, single sequences (linkers) are attached that are complementary to the single-stranded ends produced by the restriction enzyme to be used. The same enzyme is used to cut the vector, e.g., a plasmid, so that the cDNA can be incorporated for cloning.

B. Cloning vectors

The cell-based cloning of DNA fragments of different sizes is facilitated by a wide variety of vector systems. Plasmid vectors are used to clone small DNA fragments in bacteria. Their main disadvantage is that only 5–10 kb of foreign DNA can be cloned. A plasmid cloning vector that has taken up a DNA fragment (recombinant vector), e.g., pUC8 with 2.7 kb of DNA, must be distinguished from one that has not. In addition, an ampicillin resistance gene (*Amp^r*) serves to distinguish bacteria that have taken up plasmids from those that have not. Several

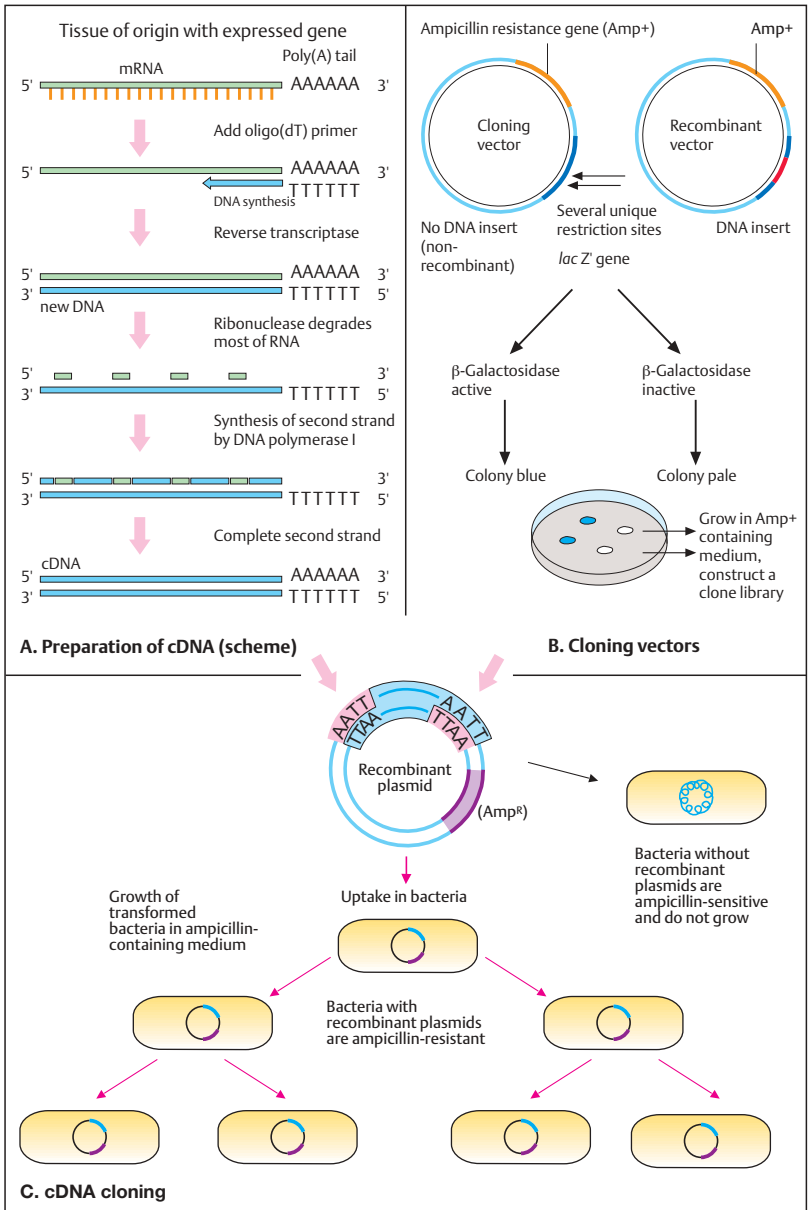
unique restriction sites in the plasmid DNA segment where a DNA fragment might be inserted serve as markers along with a marker gene, such as the *lacZ* gene. The uptake of a DNA fragment by a plasmid vector disrupts the plasmid's marker gene. Thus, in the recombinant plasmid the enzyme β -galactosidase will not be produced by the disrupted *lacZ* gene, whereas in the plasmid without a DNA insert (nonrecombinant) the enzyme is produced by the still intact *lacZ* gene. The activity of the gene and the presence or absence of the enzyme are determined by observing a difference in color of the colonies in the presence of an artificial substrate sugar. β -Galactosidase splits an artificial sugar (5-bromo-4-chloro-3-indolyl- β -D-galactopyranoside) that is similar to lactose, the natural substrate for this enzyme, into two sugar components, one of which is blue. Thus, bacterial colonies containing nonrecombinant plasmids with an intact *lacZ* gene are blue. In contrast, colonies that have taken up a recombinant vector remain pale white. The latter are grown in a medium containing ampicillin (the selectable marker for the uptake of plasmid vectors). Subsequently, a clone library can be constructed. (Figure adapted from Brown, 1999).

C. cDNA cloning

Only those bacteria become ampicillin resistant that have incorporated a recombinant plasmid. Recombinant plasmids, which contain the gene for ampicillin resistance, transform ampicillin-sensitive bacteria into ampicillin-resistant bacteria. In an ampicillin-containing medium, only those bacteria grow that contain the recombinant plasmid with the desired DNA fragment. By further replication in these bacteria, the fragment can be cloned until there is enough material to be studied. (Figures after Watson et al., 1987).

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DNA Libraries

A DNA library is a collection of DNA fragments that in their entirety represent the genome, that is, a particular gene being sought and all remaining DNA. It is the starting point for cloning a gene of unknown chromosomal location. To produce a library, the total DNA is digested with a restriction enzyme, and the resulting fragments are incorporated into vectors and replicated in bacteria. A sufficient number of clones must be present so that every segment is represented at least once. This is a question of the size of the genome being investigated and the size of the fragments. Plasmids and phages are used as vectors. For larger DNA fragments, yeast cells may be employed. There are two different types of libraries: genomic DNA and cDNA.

A. Genomic DNA library

Clones of genomic DNA are copies of DNA fragments from all of the chromosomes (1). They contain coding and noncoding sequences. Restriction enzymes are used to cleave the genomic DNA into many fragments. Here four fragments are schematically shown, containing two genes, A and B (2). These are incorporated into vectors, e.g., into phage DNA, and are replicated in bacteria. The complete collection of recombinant DNA molecules, containing all DNA sequences of a species or individual, is called a genomic library. To find a particular gene, a screening procedure is required (see B).

B. cDNA library

Unlike a genomic library, which is complete and contains coding and noncoding DNA, a cDNA library consists only of coding DNA sequences. This specificity offers considerable advantages over genomic DNA. However, it requires that mRNA be available and does not yield information about the structure of the gene. mRNA can be obtained only from cells in which the respective gene is transcribed, i.e., in which mRNA is produced (1). In eukaryotes, the RNA formed during transcription (primary transcript) undergoes splicing to form mRNA (2, see p. 50). Complementary DNA (cDNA) is formed from mRNA by the enzyme reverse transcriptase (3). The cDNA can serve as a template for synthesis of a complementary DNA strand, so that complete double-stranded DNA can be formed (cDNA clone). Its sequence corresponds to the

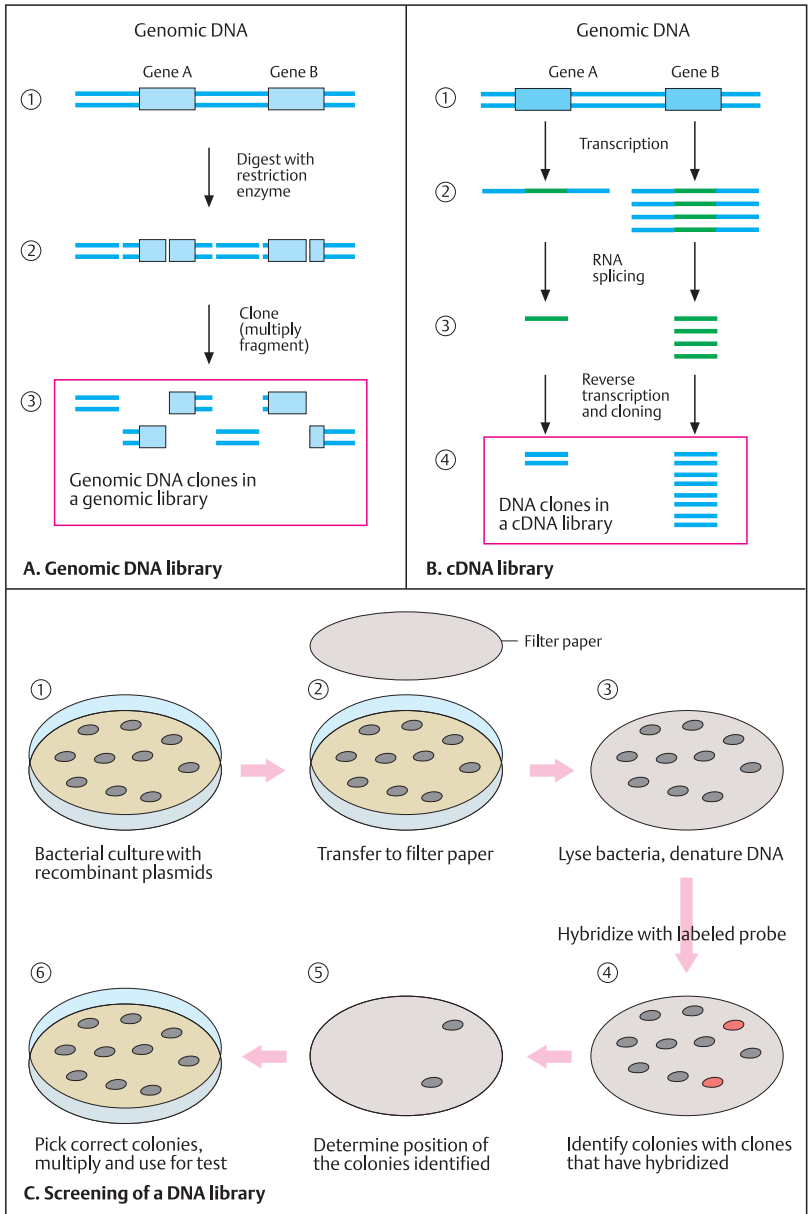
coding sequences of the gene exons. Thus it is well suited for use as a probe (cDNA probe). The subsequent steps, incorporation into a vector and replication in bacteria, correspond to those of the procedure to produce a genomic library. cDNA clones can only be won from coding regions of an active (mRNA-producing) gene; thus, the cDNA clones of different tissues differ according to genetic activity. Since cDNA clones correspond to the coding sequences of a gene (exons) and contain no noncoding sections (introns), cloned cDNA is the preferred starting material when further information about a gene product is sought by analyzing the gene. The sequence of amino acids in a protein can be determined from cloned and sequenced cDNA. Also, large amounts of a protein can be produced by having the cloned gene expressed in bacteria or yeast cells.

C. Screening of a DNA library

Bacteria that have taken up the vectors can grow on an agar-coated Petri dish, where they form colonies (1). A replica imprint of the culture is taken on a membrane (2), and the DNA that sticks to the membrane is denatured with an alkaline solution (3). DNA of the gene segment being sought can then be identified by hybridization with a radioactively (or otherwise) labeled probe (4). After hybridization, a signal appears on the membrane at the site of the gene segment (5). DNA complementary to the labeled probe is located here; its exact position in the culture corresponds to that of the signal on the membrane (6). A probe is taken from the corresponding area of the culture (5). It will contain the desired DNA segment, which can now be further replicated (cloned) in bacteria. By this means, the desired segment can be enriched and is available for subsequent studies.

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Restriction Analysis by Southern Blot Analysis

Restriction endonucleases are DNA-cleaving enzymes with defined sequences as targets (see next plate). They are often simply called restriction enzymes. Since each enzyme cleaves DNA only at its specific recognition sequence, the total DNA of an individual present in nucleated cells can be cut into pieces of manageable and defined size in a reproducible way. Individual DNA fragments can then be selected, ligated into suitable vectors, multiplied, and examined. Owing to the uneven distribution of recognition sites, the DNA fragments differ in size. A starting mixture of DNA fragments is sorted according to size. Two procedures detect target DNA or RNA fragments after they have been arranged by size in gel electrophoresis—the Southern blot hybridization for DNA (named after E. Southern who developed this method 1975) and the Northern blot hybridization for RNA (a word play on Southern, not named after a Dr. Northern). Immunoblotting (Western blot) detects proteins by an antibody-based procedure.

A. Southern blot hybridization

The analysis starts with total DNA (1). The DNA is isolated and cut with restriction enzymes (2). One of the not yet identified fragments contains the gene being sought or part of the gene. The fragments are sorted by size in a gel (usually agarose) in an electric field (electrophoresis) (3). The smaller the fragment, the faster it migrates; the larger, the slower it migrates. Next, the blot is carried out: The fragments contained in the gel are transferred to a nitrocellulose or nylon membrane (4). There the DNA is denatured (made single-stranded) with alkali and fixed to the membrane by moderate heating (-80°C) or UV cross-linkage. The sample is incubated with a probe of complementary single-stranded DNA (genomic DNA or cDNA) from the gene (5). The probe hybridizes solely with the complementary fragment being sought, and not with others (6). Since the probe is labeled with radioactive ^{32}P , the fragment being sought can be identified by placing an X-ray film on the membrane, where it appears as a black band on the film after development (autoradiogram) (6). The size, corresponding to position, is deter-

mined by running DNA fragments of known size in the electrophoresis.

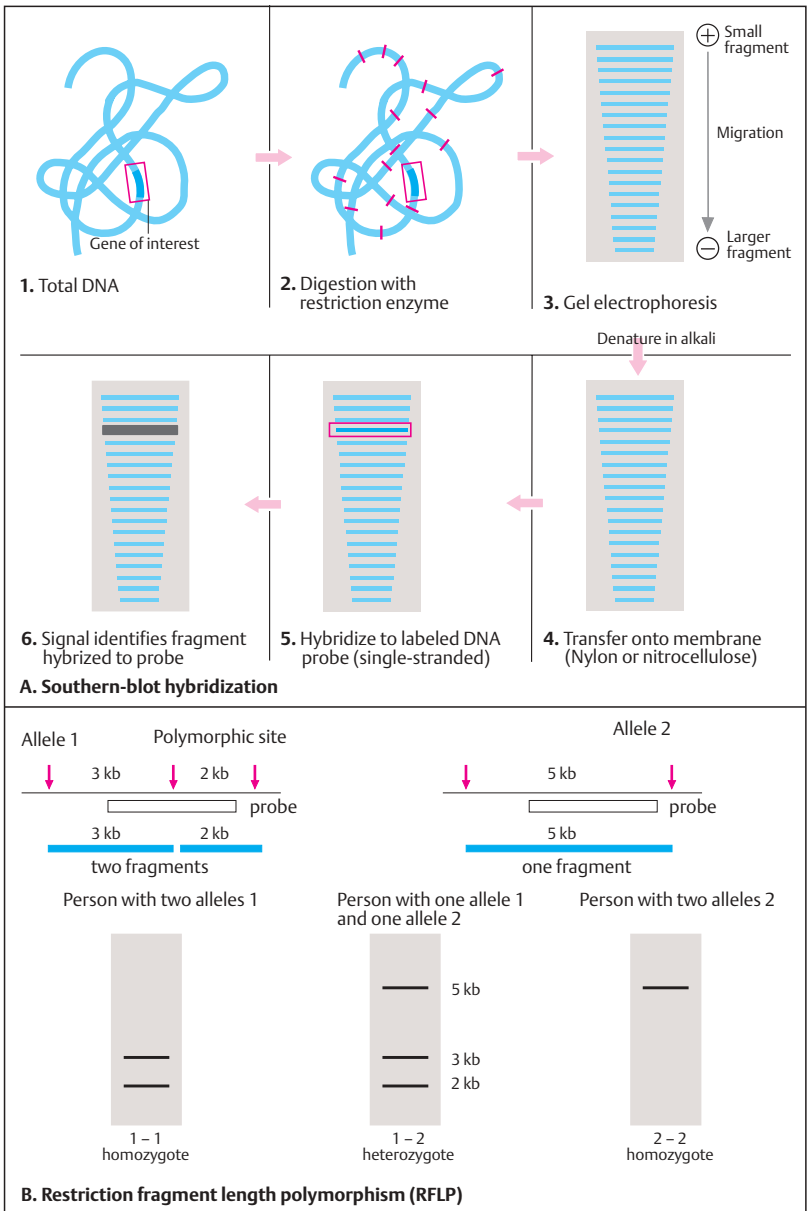
B. Restriction fragment length polymorphism (RFLP)

In about every 100 base pairs of a DNA segment, the nucleotide sequence differs in some individuals (DNA polymorphism). As a result, the recognition sequence of a restriction enzyme may be present on one chromosome but not the other. In this case the restriction fragment sizes differ at this site (restriction fragment length polymorphism, RFLP). An example is shown for two 5 kb (5000 base pair) DNA segments. In one, a restriction site in the middle is present (allele 1); in the other (allele 2) it is absent. With a Southern blot, it can be determined whether in this location an individual is homozygous 1–1 (two alleles 1, no 5 kb fragment), heterozygous 1–2 (one allele each, 1 and 2), or homozygous 2–2 (two alleles 2). If the mutation being sought lies on the chromosome carrying the 5 kb fragment, the presence of this fragment indicates presence of the mutation. The absence of this fragment would indicate that the mutation is absent.

It is important to understand that the RFLP itself is unrelated to the mutation. It simply distinguishes DNA fragments of different sizes from the same region. These can be used as markers to distinguish alleles in a segregation analysis (see p. 134). In addition to RFLPs, other types of DNA polymorphism can be detected by Southern blot hybridization, although polymerase chain reaction-based analysis of microsatellites is now used more frequently (see p. 244).

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Restriction Mapping

Restriction endonucleases (restriction enzymes) are DNA-cutting enzymes. They are obtained from bacteria, which produce the enzymes as protection from foreign DNA. A given enzyme recognizes a specific sequence of 4–8 (usually 6) nucleotides (a restriction site) where it cleaves the DNA. The sizes of the DNA fragments produced depend on the distribution of the restriction sites. More than 400 different types of restriction enzymes have been isolated.

A. DNA cleavage by restriction nucleases

The cleavage patterns (recognition sequences) of three frequently used restriction enzymes, *EcoRI*, *HindIII*, and *HpaI*, are presented. For *EcoRI* and *HindIII* the cut is “palindromic,” i.e., the cut is asymmetric around an axis on which mirror-image complementary single-stranded DNA segments arise. Each corresponds to its opposite-lying strand in the reverse direction. Therefore, they can be joined to a DNA fragment whose ends contain complementary single-stranded sequences. *HpaI* cuts both strands so that no single-stranded ends are formed. Frequently cutting and seldom cutting enzymes can be distinguished according to the frequency of occurrence of their recognition sites.

B. Examples of restriction enzymes

The recognition sequences of some restriction enzymes are shown. The names of the enzymes are derived from those of the bacteria in which they occur, e.g., *EcoRI* from *Escherichia coli* Restriction enzyme I, etc. Some enzymes have a cutting site with limited specificity. In *HindIII* it suffices that the two middle nucleotides are a pyrimidine and a purine (GTPyPuAC), and it does not matter whether the former is thymine (T) or cytosine (C), and whether the latter is adenine (A) or guanine (G). Such a recognition site occurs frequently and produces many relatively small fragments, whereas enzymes that cut very infrequently produce few and large DNA fragments.

C. Restriction fragments

In a given DNA segment, the recognition sequence of a restriction enzyme occurs irregularly. Thus, the distances between restriction

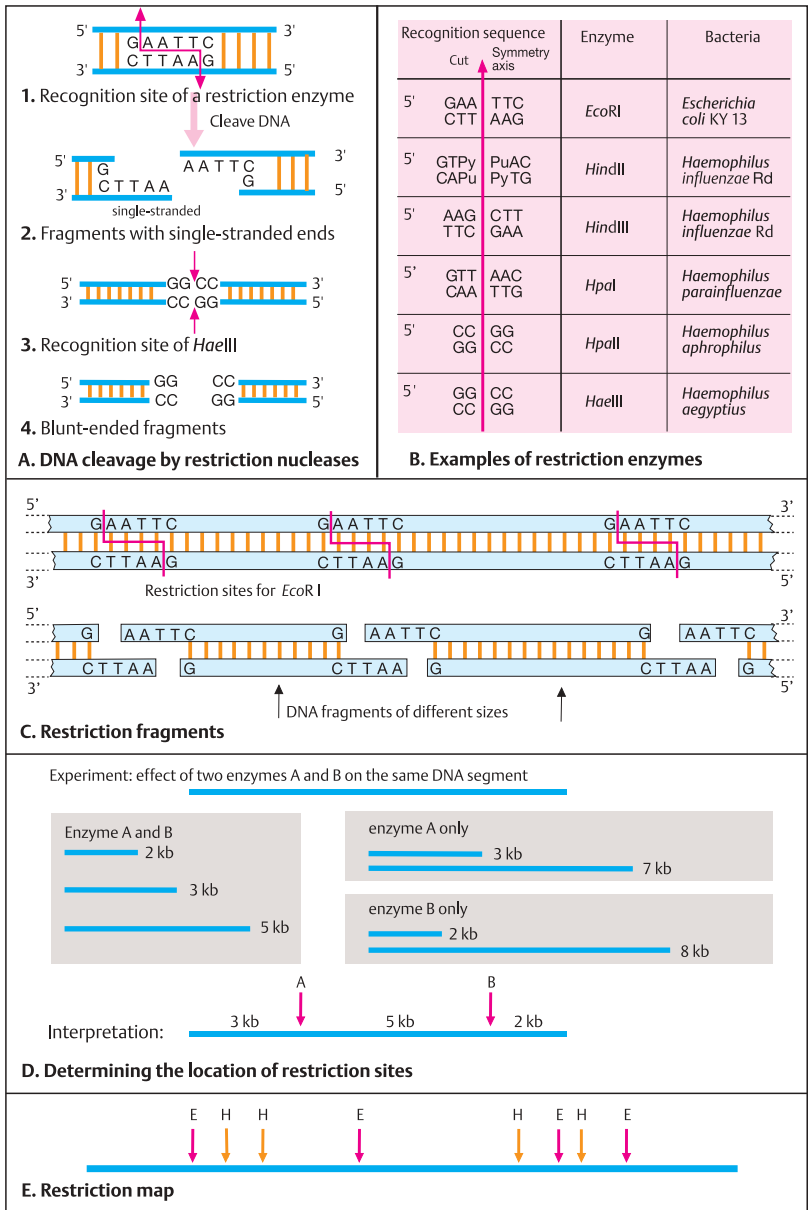
sites differ. DNA fragments of various sizes (restriction fragments) result from digestion with a restriction enzyme. A given restriction enzyme will cleave a given segment of DNA into a series of DNA fragments of characteristic sizes. This leads to a pattern that can be employed for diagnostic purposes.

D. Determination of the locations of restriction sites

Since the fragment sizes reflect the relative positions of the cutting sites, they can be used to characterize a DNA segment (restriction map). If a 10-kb DNA segment cut by two enzymes, A and B, yields three fragments, of 2 kb, 3 kb, and 5 kb, then the relative location of the cutting sites can be determined by using enzymes A and B alone in further experiments. If enzyme A yields two fragments of 3 kb and 7 kb, and enzyme B two fragments of 2 kb and 8 kb, then the two cutting sites of enzymes A and B must lie 5 kb apart. To the left of the restriction site of enzyme A are 3 kb; to the right of the restriction site of enzyme B, 2 kb (1 kb = 1000 base pairs).

E. Restriction map

A given DNA segment can be characterized by the distribution pattern of restriction sites. In the example shown, a DNA segment is characterized by the distribution of the cutting sites for enzymes E (*EcoRI*) and H (*HindIII*). The individual sites are separated by intervals defined by the size of the fragments after digestion with the enzyme. A restriction map is a linear sequence of restriction sites at defined intervals along the DNA. Restriction mapping is of considerable importance in medical genetics and evolutionary research.



DNA Amplification by Polymerase Chain Reaction (PCR)

The introduction of cell-free methods for multiplying DNA fragments of defined origin (DNA amplification) in 1985 ushered in a new era in molecular genetics (the principle of PCR is contained in earlier publications). This fundamental technology has spread dramatically with the development of automated equipment used in basic and applied research.

A. Polymerase chain reaction (PCR)

PCR is a cell-free, rapid, and sensitive method for cloning DNA fragments. A standard reaction and a wide variety of PCR-based methods have been developed to assay for polymorphisms and mutations. Standard PCR is an *in vitro* procedure for amplifying defined target DNA sequences, even from very small amounts of material or material of ancient origin. Selective amplification requires some prior information about DNA sequences flanking the target DNA. Based on this information, two oligonucleotide primers of about 15–25 base pairs length are designed. The primers are complementary to sequences outside the 3' ends of the target site and bind specifically to these.

PCR is a chain reaction because newly synthesized DNA strands act as templates for further DNA synthesis for about 25–35 subsequent cycles. Theoretically each cycle doubles the amount of DNA amplified. At the end, at least 10^5 copies of the specific target sequence are present. This can be visualized as a distinct band of a specific size after gel electrophoresis. Each cycle, involving three precisely time-controlled and temperature-controlled reactions in automated thermal cyclers, takes about 1–5 min. The three steps in each cycle are (1) denaturation of double-stranded DNA, at about 93–95°C for human DNA, (2) primer annealing at about 50–70°C depending on the expected melting temperature of the duplex DNA, and (3) DNA synthesis using heat-stable DNA polymerase (from microorganisms living in hot springs, such as *Thermophilus aquaticus*, Taq polymerase), typically at about 70–75°C. At each subsequent cycle the template (shown in blue) and the DNA newly synthesized during the preceding cycle (shown in red) act as tem-

plates for another round of synthesis. The first cycle results in newly synthesized DNA of varied lengths (shown with an arrow) at the 3' ends because synthesis is continued beyond the target sequences. The same happens during subsequent cycles, but the variable strands are rapidly outnumbered by new DNA of fixed length at both ends because synthesis cannot proceed past the terminus of the primer at the opposite template DNA.

B. cDNA amplification and RT-PCR

A partially known amino acid sequence of a polypeptide can be used to obtain the sequence information required for PCR. From its mRNA one can derive cDNA (see complementary DNA, p. 58) and determine the sequence of the sense and the antisense strand to prepare appropriate oligonucleotide primers (1). When different RNAs are available in small amounts, rapid PCR-based methods are employed to amplify cDNA from different exons of a gene. cDNA is obtained by reverse transcriptase from mRNA, which is then removed by alkaline hydrolysis (2). After a complementary new DNA strand has been synthesized, the DNA can be amplified by PCR (3). Reverse transcriptase PCR (RT-PCR) can be used when the known exon sequences are widely separated within a gene. With rapid amplification of cDNA ends (RACE-PCR), the 5' and 3' end sequences can be isolated from cDNA.

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